P.6-15 Use phasors and cosine reference.

$$\bar{H} = \bar{a}_\phi H_0, \quad \bar{H}_1 = \bar{a}_\phi 10^{-4} e^{-j\pi/2}, \quad \bar{H}_2 = \bar{a}_\phi 2\times 10^{-4} e^{j\alpha}.$$

$$\longrightarrow H_0 = 10^{-4}(-j + 2e^{j\alpha})$$

$$= 10^{-4}\left[2\cos\alpha + j(2\sin\alpha - 1)\right].$$

$$2\sin\alpha - 1 = 0 \longrightarrow \alpha = 30^\circ, \text{ or } \pi/6 \text{ (rad.)}$$

$$H_0 = 2\times 10^{-4}\cos 30^\circ = 1.73\times 10^{-4} \text{ (A/m)}.$$

P. 6-17 See Section 10-2, pp. 428-429, Eqs. (10-6) and (10-7).

P. 6-18 a) $k = \dfrac{\omega}{c} = \dfrac{2\pi(60\times 10^6)}{3\times 10^8} = 0.4\pi$ (rad/m).

b)

$$\bar{H} = \frac{1}{-j\omega\mu_0}\bar{\nabla}\times\bar{E} = \frac{j}{\omega\mu_0}\begin{vmatrix} \bar{a}_r & \bar{a}_\phi & \bar{a}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ E_r & 0 & 0 \end{vmatrix}$$

$$= \frac{j}{\omega\mu_0}\bar{a}_\phi \frac{\partial E_r}{\partial z} = \bar{a}_\phi \frac{j}{\omega\mu_0}(-jk)\frac{E_0}{r}e^{-jkz}$$

$$= \bar{a}_\phi \frac{k}{\omega\mu_0}\frac{E_0}{r}e^{-jkz} = \bar{a}_\phi \frac{E_0}{120\pi r}e^{-j0.4\pi z} \quad \text{(A/m)}.$$

c) $$\bar{J}_s\Big|_{r=a} = \bar{a}_z H_\phi\Big|_{r=a} = \bar{a}_z \frac{E_0}{120\pi a}e^{-j0.4\pi z} \quad \text{(A/m)}.$$

$$\bar{J}_s\Big|_{r=b} = -\bar{a}_z H_\phi\Big|_{r=b} = -\bar{a}_z \frac{E_0}{120\pi b}e^{-j0.4\pi z} \quad \text{(A/m)}.$$

P.6-19 $k = \frac{\omega}{c} = \frac{2\pi \times 10^9}{3\times 10^8} = \frac{20\pi}{3}$ (rad/m).

$$\bar{H} = \frac{j}{\omega\mu_0}\bar{\nabla}\times\bar{E}.$$

In phasor form: $\bar{E} = \bar{a}_\theta \frac{10^{-3}}{R}\sin\theta\, e^{-jkR}$.

From Eq. (2-99):
$$\bar{H} = \frac{j}{\omega\mu_0}\frac{1}{R^2\sin\theta}\begin{vmatrix} \bar{a}_R & \bar{a}_\theta R & \bar{a}_\phi R\sin\theta \\ \frac{\partial}{\partial R} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ 0 & 10^{-3}\sin\theta\cdot e^{-jkR} & 0 \end{vmatrix}$$
$$= \bar{a}_\phi \frac{10^{-3}}{\eta_0 R}\sin\theta\cdot e^{-jkR}$$

In instantaneous form:
$$\bar{H}(R,\theta;t) = \bar{a}_\phi \frac{10^{-3}}{120\pi R}\sin\theta\cos(2\pi 10^9 t - 20\pi R/3)\ \text{(A/m)}.$$

P.6-20 In phasor form:
$$\bar{E} = \bar{a}_y\, 0.1\sin(10\pi x)\, e^{-j\beta z}. \qquad ①$$
$$\bar{H} = -\frac{1}{j\omega\mu_0}\bar{\nabla}\times\bar{E}$$
$$= \frac{j}{\omega\mu_0}\left[\bar{a}_x\, j0.1\beta\sin(10\pi x) + \bar{a}_z\, 0.1(10\pi)\cos(10\pi x)\right]e^{-j\beta z} \qquad ②$$
$$\bar{E} = \frac{1}{j\omega\epsilon_0}\bar{\nabla}\times\bar{H}$$
$$= \bar{a}_y \frac{0.1}{\omega^2\mu_0\epsilon_0}\left[(10\pi)^2+\beta^2\right]\sin(10\pi x)\, e^{-j\beta z} \qquad ③$$

Equating ① and ③: $(10\pi)^2 + \beta^2 = \omega^2\mu_0\epsilon_0 = 400\pi^2$.
$$\longrightarrow \beta = \sqrt{300}\,\pi = 54.4\ \text{(rad/m)}.$$

From ②: $\bar{H}(x,z;t) = \mathcal{R}e\left(\bar{H}e^{j\omega t}\right)$
$$= -\bar{a}_x\, 2.30\times10^{-4}\sin(10\pi x)\cos(6\pi 10^9 t - 54.4z)$$
$$- \bar{a}_z\, 1.33\times10^{-4}\cos(10\pi x)\sin(6\pi 10^9 t - 54.4z)\ \text{(A/m)}$$

P. 6-21 $\quad \bar{H}(x,z;t) = \bar{a}_y 2\cos(15\pi x)\sin(6\pi 10^9 t - \beta z)$ (A/m).

Phasor with sine reference:

$$\bar{H} = \bar{a}_y 2\cos(15\pi x)\cdot e^{-j\beta z} \qquad ①$$

$$\bar{E} = \frac{1}{j\omega\epsilon_0}\bar{\nabla}\times\bar{H}$$

$$= \frac{1}{j\omega\epsilon_0} 2\left[\bar{a}_x j\beta\cos(15\pi x)e^{-j\beta z} - \bar{a}_z 15\pi\sin(15\pi x)e^{-j\beta z}\right]. \qquad ②$$

$$\bar{H} = -\frac{1}{j\omega\mu_0}\bar{\nabla}\times\bar{E}$$

$$= \frac{2}{\omega^2\mu_0\epsilon_0}\left[\bar{a}_y\left(-\frac{\partial E_z}{\partial x} + \frac{\partial E_x}{\partial z}\right)\right]$$

$$= \bar{a}_y \frac{2}{\omega^2\mu_0\epsilon_0}\left[(15\pi)^2 + \beta^2\right]\cos(15\pi x)\cdot e^{-j\beta z} \qquad ③$$

Comparing ① and ③, we require

$$(15\pi)^2 + \beta^2 = \omega^2\mu_0\epsilon_0 = \frac{(6\pi 10^9)^2}{c^2}$$

$$= \frac{(6\pi 10^9)^2}{(3\times 10^8)^2} = 400\pi^2.$$

$$\longrightarrow \beta = 13.2\pi = 41.6 \text{ (rad/m)}.$$

From ②, we have

$$\bar{E}(x,z;t) = \mathcal{I}_m(\bar{E}e^{j\omega t})$$

$$= \bar{a}_x 496\cos(15\pi x)\sin(6\pi 10^9 t - 41.6z)$$

$$+ \bar{a}_z 565\sin(15\pi x)\cos(6\pi 10^9 t - 41.6z) \quad \text{(V/m)}.$$

Chapter 7

Plane Electromagnetic Waves

P. 7-1 a) In a source-free conducting medium with constitutive parameters ϵ, μ, and σ,

$$\text{Eq. (7-62):}\quad \bar{\nabla}\times\bar{H} = \bar{J} + \frac{\partial\bar{D}}{\partial t}$$

$$= \sigma\bar{E} + \epsilon\frac{\partial\bar{E}}{\partial t}. \qquad ①$$

$$\text{Eqs. (5-16a) \& (7-61):}\quad \bar{\nabla}\times\bar{\nabla}\times\bar{E} = \bar{\nabla}\bar{\nabla}\cdot\bar{E} - \bar{\nabla}^2\bar{E}$$

$$= -\mu\frac{\partial}{\partial t}(\bar{\nabla}\times\bar{H}). \qquad ②$$

Substituting ① in ② and noting that $\bar{\nabla}\cdot\bar{E} = 0$, we obtain the wave equation in dissipative media:

$$\bar{\nabla}^2\bar{E} - \mu\sigma\frac{\partial\bar{E}}{\partial t} - \mu\epsilon\frac{\partial^2\bar{E}}{\partial t^2} = 0. \qquad ③$$

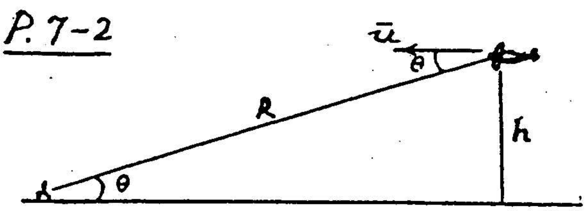
Similarly for $\bar{H}$.

b) For time-harmonic fields: $\frac{\partial}{\partial t} \to (j\omega)$ and $\frac{\partial^2}{\partial t^2} \to (-\omega^2)$.

Wave equation ③ converts to Helmholtz's equation:

$$\bar{\nabla}^2\bar{E} - j\omega\mu\sigma\bar{E} + k^2\bar{E} = 0, \quad \text{where } k = \omega\sqrt{\mu\epsilon}.$$

P. 7-2



$$\Delta t = \frac{2R}{c} = 0.3\times10^{-3}\ (\text{s}).$$

$$R = \frac{\Delta t}{2}c = \frac{0.3\times10^{-3}}{2}\times3\times10^{8}$$

$$= 45\times10^{3}\ (\text{m}),$$

or 45 (km).

$$h = R\sin\theta = 45\times10^{3}\sin 15.5^\circ = 12\times10^{3}(\text{m}), \text{ or } 12\ (\text{km}).$$

$$\Delta f = 2f\left(\frac{u}{c}\right)\cos 15.5^\circ.$$

$$\longrightarrow u = \frac{c\,\Delta f}{2f\cos 15.5^\circ} = 410.8\ (\text{m/s}), \text{ or about } 1.2 \text{ Mach}.$$

P.7-3 Assume that $\bar{H}(\bar{R})$ has the form:

$$\bar{H}(\bar{R}) = \bar{H}_0 e^{-jk\bar{a}_k\cdot\bar{R}}.$$

Then, from Eq. (6-80b),

$$\bar{E}(\bar{R}) = \frac{1}{j\omega\epsilon}\nabla\times\bar{H}(\bar{R})$$

$$= \frac{1}{j\omega\epsilon}(-jk)\bar{a}_k\times\bar{H}(\bar{R})$$

$$= -\frac{1}{\omega\epsilon}(\omega\sqrt{\mu\epsilon})\,\bar{a}_k\times\bar{H}(\bar{R}),$$

or,

$$\bar{E}(\bar{R}) = -\eta\,\bar{a}_k\times\bar{H}(\bar{R}).$$

P.7-4 $\bar{H} = \bar{a}_z\, 4\times10^{-6}\cos(10^7\pi t - k_0 y + \frac{\pi}{4})$ (A/m).

a) $k_0 = \omega\sqrt{\mu_0\epsilon_0} = \frac{10^7\pi}{3\times10^8} = \frac{\pi}{30} = 0.105$ (rad/m).

$\lambda = 2\pi/k_0 = 60$ (m).

At $t = 3\times10^{-3}$ (s), we require the argument of cosine in $\bar{H}$:

$$10^7\pi(3\times10^{-3}) - \frac{\pi}{30}y + \frac{\pi}{4} = \pm n\pi + \frac{\pi}{2}, \quad n = 0, 1, 2, \cdots$$

$$\longrightarrow\ y = \pm30n - 7.5\ \text{(m)} = 22.5 \pm n\lambda/2\ \text{(m)}.$$

b) Use phasors with cosine reference:

$$\bar{H} = \bar{a}_z\, 4\times10^{-6}\, e^{j(-k_0 y + \pi/4)} \quad \text{(A/m)}.$$

From the result of Problem P.7-3,

$$\bar{E} = -\eta_0\,\bar{a}_y\times\bar{a}_z\, 4\times10^{-6} e^{j(-k_0 y + \pi/4)}$$

$$= -\bar{a}_x\, 4\times10^{-6}\eta_0\, e^{j(-0.105y + \pi/4)}$$

$$= -\bar{a}_x\, 1.51\times10^{-3} e^{j(-0.105y + \pi/4)} \quad \text{(V/m)}.$$

The instantaneous expression for $\bar{E}$ is:

$$\bar{E}(y,t) = -\bar{a}_x\, 1.51\cos(10^7\pi t - 0.105y + \pi/4) \quad \text{(mV/m)}.$$

P.7-5 Use phasors with cosine reference.

$$\bar{E}(z) = \bar{a}_x 2 e^{-jz/\sqrt{3}} + \bar{a}_y j e^{-jz/\sqrt{3}} \quad (\text{V/m}).$$

a) $\omega = 10^8$ (rad/s) $\longrightarrow$ $f = 10^8/2\pi = 1.59 \times 10^7$ (Hz),
$\beta = 1/\sqrt{3}$ (rad/m) $\longrightarrow$ $\lambda = 2\pi/\beta = 2\sqrt{3}\,\pi$ (m).

b) $u = \frac{c}{\sqrt{\epsilon_r}} = \frac{\omega}{\beta} \longrightarrow \epsilon_r = \left(\frac{\beta c}{\omega}\right)^2 = 3.$

c) Left-hand elliptically polarized.

d) $\eta = \sqrt{\frac{\mu}{\epsilon}} = \frac{120\pi}{\sqrt{\epsilon_r}} = \frac{120\pi}{\sqrt{3}}$ (Ω).

$$\bar{H} = \frac{1}{\eta}\bar{a}_z \times \bar{E} = \frac{\sqrt{3}}{120\pi}(\bar{a}_y 2 e^{-jz/\sqrt{3}} - \bar{a}_x j e^{-jz/\sqrt{3}}),$$

$$\bar{H}(z,t) = \frac{\sqrt{3}}{120\pi}\left[\bar{a}_x \sin(10^8 t - z/\sqrt{3}) + \bar{a}_y 2\cos(10^8 t - z/\sqrt{3})\right] \quad (\text{A/m}).$$

P.7-6 Let $\alpha = \omega t - kz$.

$$\bar{E} = \bar{a}_x E_{10} \sin\alpha + \bar{a}_y E_{20} \sin(\alpha + \psi)$$
$$= \bar{a}_x E_x + \bar{a}_y E_y .$$

$$\frac{E_x}{E_{10}} = \sin\alpha, \quad \frac{E_y}{E_{20}} = \sin(\alpha+\psi)$$
$$= \sin\alpha\cos\psi + \cos\alpha\sin\psi$$
$$= \frac{E_x}{E_{10}}\cos\psi + \sqrt{1-\left(\frac{E_x}{E_{10}}\right)^2}\sin\psi .$$

$$\left(\frac{E_y}{E_{20}} - \frac{E_x}{E_{10}}\cos\psi\right)^2 = \left[1 - \left(\frac{E_x}{E_{10}}\right)^2\right]\sin^2\psi .$$

Rearranging:

$$\left(\frac{E_y}{E_{20}\sin\psi}\right)^2 + \left(\frac{E_x}{E_{10}\sin\psi}\right)^2 - 2\frac{E_x E_y}{E_{10}E_{20}}\frac{\cos\psi}{\sin^2\psi} = 1,$$

which is the equation of an ellipse in E_x–E_y plane.

P.7-7 Given: $f = 3\times10^9$ (Hz), $\epsilon_r = 2.5$,

$$\tan\delta_c = \frac{\epsilon''}{\epsilon'} = 0.05.$$

a) Eq.(7-47): $\alpha = \frac{\omega\epsilon''}{2}\sqrt{\frac{\mu}{\epsilon'}} = \frac{\omega}{2}\left(\frac{\epsilon''}{\epsilon'}\right)\frac{\sqrt{\epsilon_r}}{c} = 2.48$ (Np/m).

$$e^{-\alpha x} = \frac{1}{2} \longrightarrow x = \frac{1}{\alpha}\ln 2 = 0.279 \text{ (m)}.$$

b) Eq.(7-49): $\eta_c = \frac{1}{\sqrt{\epsilon_r}}\sqrt{\frac{\mu_0}{\epsilon_0}}\left(1 + j\frac{\epsilon''}{2\epsilon'}\right) = 238\angle 1.43^\circ$ (Ω)
$= 238\angle 0.008\pi$ (Ω).

Eq.(7-48): $\beta = \omega\sqrt{\mu\epsilon}\left[1 + \frac{1}{8}\left(\frac{\epsilon''}{\epsilon'}\right)^2\right] = 31.6\pi$ (rad/m).

$$\lambda = \frac{2\pi}{\beta} = 0.063 \text{ (m)}.$$

$$u_p = \frac{\omega}{\beta} = 1.897\times10^8 \text{ (m/s)}.$$

$$u_g = \frac{1}{d\beta/d\omega} = \frac{c}{\sqrt{\epsilon_r}}\left[1 + \frac{1}{8}\left(\frac{\epsilon''}{\epsilon'}\right)^2\right] = 1.898\times10^8 \text{ (m/s)}.$$

c) At $x=0$, $\bar{E} = \bar{a}_y e^{j\pi/3}$, $\bar{H} = \frac{1}{\eta_c}\bar{a}_x\times\bar{E} = \bar{a}_z 0.210 e^{j(\frac{\pi}{3} - 0.008\pi)}$

$$\bar{H}(x,t) = \bar{a}_z 0.210 e^{-2.48x}\sin(6\pi 10^9 t - 31.6\pi x + 0.325\pi) \text{ (A/m)}.$$

P.7-8 Both copper and brass are good conductors,

$\left(\frac{\sigma}{\omega\epsilon}\right)^2 \gg 1$: $\alpha = \sqrt{\pi f\mu\sigma}$, $\delta = \frac{1}{\alpha}$, $\eta_c = (1+j)\frac{\alpha}{\sigma}$.

a) $f = 1$ (MHz).

	η_c (Ω)	α (Np/m)	α (dB/m)	δ (m)
Copper	$2.61(1+j)\times10^{-4}$	1.51×10^4	1.31×10^5	6.61×10^{-5}
Brass	$4.98(1+j)\times10^{-4}$	0.79×10^4	0.69×10^5	12.6×10^{-5}

b) $f = 1$ (GHz).

	η_c (Ω)	α (Np/m)	α (dB/m)	δ (m)
Copper	$8.25(1+j)\times10^{-3}$	4.79×10^5	4.16×10^6	2.09×10^{-6}
Brass	$1.58(1+j)\times10^{-3}$	2.51×10^5	2.18×10^6	3.99×10^{-6}

P.7-9 a) $\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} \longrightarrow \sigma = \frac{1}{\pi f \mu \delta^2} = 0.99\times10^5\ (S/m)$.

b) At $f = 10^9\ (Hz)$, $\alpha = \sqrt{\pi f \mu \sigma} = 1.98\times10^4\ (Np/m)$.

$$20 \log_{10} e^{-\alpha z} = -30\ (dB) \longrightarrow z = \frac{1.5}{\alpha \log_{10} e} = 1.75\times10^{-4}\ (m) = 0.175\ (mm).$$

P.7-10 $\mathcal{P}_{av} = |E|^2/2\eta_0 = 10^{-2}\ (W/cm^2)$.

a) $|E| = \sqrt{0.02\eta_0} = 2.75\ (V/cm) = 275\ (V/m)$,
$|H| = |E|/\eta_0 = 7.28\times10^{-3}\ (A/cm) = 0.728\ (A/m)$.

b) $\mathcal{P}_{av} = |E|^2/2\eta_0 = 1300\ (W/m^2)$.
$|E| = 990\ (V/m)$, $\quad |H| = 2.63\ (A/m)$.

P.7-11 Assume circularly polarized plane wave:

$$\bar{E}(z,t) = \bar{a}_x E_0 \cos(\omega t - kz + \phi) + \bar{a}_y E_0 \sin(\omega t - kz + \phi),$$
$$\bar{H}(z,t) = \bar{a}_y \frac{E_0}{\eta} \cos(\omega t - kz + \phi) - \bar{a}_x \frac{E_0}{\eta} \sin(\omega t - kz + \phi).$$

Poynting vector, $\bar{\mathcal{P}} = \bar{E}\times\bar{H} = \bar{a}_z \frac{E_0^2}{\eta}\left[\cos^2(\omega t - kz + \phi) + \sin^2(\omega t - kz + \phi)\right]$
$= \bar{a}_z \frac{E_0^2}{\eta}$, a constant independent of t and z.

P.7-12 $\bar{E} = \bar{a}_\theta E_\theta + \bar{a}_\phi E_\phi$,

$$\bar{H} = \frac{1}{\eta}\bar{a}_R \times \bar{E} = \frac{1}{\eta}(\bar{a}_\phi E_\theta - \bar{a}_\theta E_\phi).$$
$$\bar{\mathcal{P}}_{av} = \frac{1}{2}\mathcal{R}e(\bar{E}\times\bar{H}^*) = \bar{a}_z \frac{1}{2\eta}(|E_\theta|^2 + |E_\phi|^2).$$

P.7-13 From Gauss's law: $\bar{E} = \bar{a}_r \frac{\rho}{2\pi\epsilon r}$, where ρ is the line charge density on the inner conductor.

$$V_0 = -\int_b^a \bar{E}\cdot d\bar{r} = \frac{\rho}{2\pi\epsilon}\ln\left(\frac{b}{a}\right) \longrightarrow \bar{E} = \bar{a}_r \frac{V_0}{r\ln(b/a)}.$$

From Ampère's circuital law, $\bar{H} = \bar{a}_\phi \frac{I}{2\pi r}$.

Poynting vector, $\bar{\mathcal{P}} = \bar{E}\times\bar{H} = \bar{a}_z \frac{V_0 I}{2\pi r^2 \ln(b/a)}$.

Power transmitted over cross-sectional area:

$$P = \int_S \bar{\mathcal{P}}\cdot d\bar{s} = \frac{V_0 I}{2\pi\ln(b/a)}\int_0^{2\pi}\int_a^b \left(\frac{1}{r^2}\right) r\,dr\,d\phi = V_0 I.$$

P.7-14 $\bar{E}_i(x,t) = \bar{a}_y 50 \sin(10^8 t - \beta x)$ (V/m).

Use phasors with a sine reference.

$$\bar{E}_i(x) = \bar{a}_y 50 e^{-j\beta_1 x}$$

For air (medium 1): $\beta_1 = \frac{\omega}{c} = \frac{10^8}{3\times10^8} = \frac{1}{3}$ (rad/m)

$\eta_1 = \eta_0 = 120\pi$ (Ω)

For lossless medium 2: $\beta_2 = \omega\sqrt{\mu_2\epsilon_2} = \frac{\omega}{c}\sqrt{\mu_{r_2}\epsilon_{r_2}} = \frac{4}{3}$ (rad/m).

$\eta_2 = \sqrt{\frac{\mu_0\mu_{r_2}}{\epsilon_0\epsilon_{r_2}}} = 2\eta_0 = 240\pi$ (Ω).

Eq.(7-25): $\bar{H}_i(x) = \frac{1}{\eta_0}\bar{a}_x \times \bar{E}_i = \bar{a}_z \frac{1}{\eta_0} 50 e^{-jx/3} = \bar{a}_z \frac{1}{2.4\pi} e^{-jx/3}$.

a) $$\frac{E_{r0}}{E_{i0}} = \Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{2-1}{2+1} = \frac{1}{3}.$$

$\bar{E}_r = \bar{a}_y \frac{50}{3} e^{jx/3} \longrightarrow \bar{E}_r(x,t) = \bar{a}_y \frac{50}{3} \sin(10^8 t + x/3)$ (V/m).

$\bar{H}_r = \frac{1}{\eta_0}(-\bar{a}_x)\times\bar{E}_r \longrightarrow \bar{H}_r(x,t) = -\bar{a}_z \frac{1}{7.2\pi} \sin(10^8 t + x/3)$ (A/m).

b) $\Gamma = \frac{1}{3}$, $\tau = 1 + \Gamma = \frac{4}{3}$.

$$S = \frac{1+\Gamma}{1-\Gamma} = 2.$$

c) $\bar{E}_t = (\tau E_{i0})\bar{a}_y e^{-j\beta_2 x} \longrightarrow \bar{E}_t(x,t) = \bar{a}_y \frac{200}{3} \sin(10^8 t - 4x/3)$ (V/m).

$\bar{H}_t = \frac{1}{\eta_2}\bar{a}_x \times \bar{E}_t . \longrightarrow \bar{H}_t(x,t) = \bar{a}_z \frac{1}{3.6\pi} \sin(10^8 t - 4x/3)$ (A/m).

P.7-15 $\frac{\sigma}{\omega\epsilon} = \frac{4}{10^8 \times 72 \times \frac{1}{36\pi}\times 10^{-9}} = 20\pi \gg 1$ · (Good conductor).

a) Skin depth $\delta = \frac{1}{\sqrt{\pi f \mu_0 \sigma}} = 0.063$ (m) $= 6.3$ (cm) $= \frac{1}{\alpha} = \frac{1}{15.85}$.

$\eta_c = (1+j)\frac{\alpha}{\sigma} = 3.96(1+j) = 5.60 e^{j\pi/4}$ (Ω).

b) $\bar{H}(z,t) = \bar{a}_y 0.3 e^{-15.85z}\cos(10^8 t - 15.85z)$ (A/m).

$\bar{E}(z) = -\eta_c \bar{a}_z \times \bar{H}(z) \longrightarrow \bar{E}(z,t) = \bar{a}_x 1.68 e^{-15.85z}\cos(10^8 t - 15.85z + \frac{\pi}{4})$ (V/m).

c) $\bar{\mathcal{P}}_{av} = \frac{1}{2}\mathcal{R}e(\bar{E}\times\bar{H}^*) = \bar{a}_z \frac{1.68\times0.3}{2} e^{-31.7z}\cos\frac{\pi}{4} = \bar{a}_z 0.178 e^{-31.7z}$ (W/m²).

P.7-16 a) $\Gamma = \frac{E_{ro}}{E_{io}} = \frac{-\eta_1 H_{ro}}{\eta_1 H_{io}} = \frac{\eta_2-\eta_1}{\eta_2+\eta_1}$.

$\longrightarrow \frac{H_{ro}}{H_{io}} = -\Gamma = \frac{\eta_1-\eta_2}{\eta_1+\eta_2}$.

b) $\tau = \frac{E_{to}}{E_{io}} = \frac{\eta_2 H_{to}}{\eta_1 H_{io}} = \frac{2\eta_2}{\eta_2+\eta_1}$.

$\longrightarrow \frac{H_{to}}{H_{io}} = \frac{\eta_1}{\eta_2}\tau = \frac{2\eta_1}{\eta_2+\eta_1}$.

P.7-17 Given $\bar{E}_i = E_0(\bar{a}_x - j\bar{a}_y)e^{-j\beta z}$.

a) Assume reflected $\bar{E}_r(z) = (\bar{a}_x E_{rx} + \bar{a}_y E_{ry})e^{j\beta z}$.

Boundary condition at $z=0$: $\bar{E}_i(0) + \bar{E}_r(0) = 0$.

$\longrightarrow \bar{E}_r(z) = E_0(-\bar{a}_x + j\bar{a}_y)e^{j\beta z}$, a left-hand circularly polarized wave in $-z$ direction.

b) $\bar{a}_{n2} \times (\bar{H}_1 - \bar{H}_2) = \bar{J}_s$. $\longrightarrow -\bar{a}_z \times [\bar{H}_i(0) + \bar{H}_r(0)] = \bar{J}_s$. ($\bar{H}_2 = 0$ in perfect conductor.)

$\bar{H}_i(0) = \frac{1}{\eta_0}\bar{a}_z \times \bar{E}_i(0) = \frac{E_0}{\eta_0}(j\bar{a}_x + \bar{a}_y)$, $\bar{H}_r(0) = \frac{1}{\eta_0}(-\bar{a}_z)\times\bar{E}_r(0) = \frac{E_0}{\eta_0}(j\bar{a}_x + \bar{a}_y)$.

$\bar{H}_1(0) = \bar{H}_i(0) + \bar{H}_r(0) = \frac{2E_0}{\eta_0}(j\bar{a}_x + \bar{a}_y)$,

$\bar{J}_s = -\bar{a}_z \times \bar{H}_1(0) = \frac{2E_0}{\eta_0}(\bar{a}_x - j\bar{a}_y)$.

c) $\bar{E}_1(z,t) = \mathcal{R}e\left[\bar{E}_i(z) + \bar{E}_r(z)\right]e^{j\omega t}$

$= \mathcal{R}e\, E_0\left[(\bar{a}_x - j\bar{a}_y)e^{-j\beta z} + (-\bar{a}_x + j\bar{a}_y)e^{j\beta z}\right]e^{j\omega t}$

$= \mathcal{R}e\, E_0\left[-2j(\bar{a}_x - j\bar{a}_y)\sin\beta z\right]e^{j\omega t}$

$= 2E_0 \sin\beta z\,(\bar{a}_x \sin\omega t - \bar{a}_y \cos\omega t)$.

P.7-18 For normal incidence:

$1 + \Gamma = \tau$, where $|\Gamma| \le 1$.

If $|\tau| = |\Gamma|$: $\Gamma < 0$, and $\eta_1 - \eta_2 = 2\eta_2$.

$\longrightarrow \eta_1 = 3\eta_2 \longrightarrow |\Gamma| = \frac{1}{2}$.

$S = \frac{1+|\Gamma|}{1-|\Gamma|} = 3. \longrightarrow S_{dB} = 20\log_{10}3 = 9.54$ (dB).

P. 7-19 $\bar{E}_i(z) = \bar{a}_x 10 e^{-j6z}$ (V/m).

$\longrightarrow \bar{H}_i(z) = \bar{a}_y \frac{10}{377} e^{-j6z}$ (A/m).

In air: $\beta_1 = 6 = \frac{\omega}{c}. \longrightarrow \omega = 6c = 1.8\times10^9$ (rad/s).

In lossy medium: $\epsilon_{r2} = 2.25$, $\tan\delta = \frac{\epsilon''}{\epsilon'} = 0.3$.

Eq. (7-47): $\alpha_2 = \frac{\omega\epsilon''}{2}\sqrt{\frac{\mu}{\epsilon'}} = \frac{\omega}{2}\left(\frac{\epsilon''}{\epsilon'}\right)\frac{\sqrt{\epsilon_{r2}}}{c} = 1.35$ (Np/m).

Eq. (7-48): $\beta_2 = \omega\sqrt{\mu\epsilon'}\left[1 + \frac{1}{8}\left(\frac{\epsilon''}{\epsilon'}\right)^2\right] = 9.10$ (rad/m).

Eq. (7-49): $\eta_2 = \sqrt{\frac{\mu'}{\epsilon'}}\left(1 + j\frac{\epsilon''}{2\epsilon'}\right) = \frac{377}{\sqrt{2.25}}\left(1 + j\frac{0.3}{2}\right) = 254 e^{j8.5°}$ (Ω).

$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = 0.208 e^{j159.9°}$

$\tau = 1 + \Gamma = 0.805 + j0.072 = 0.808 e^{j5.1°}$

a) $\bar{E}_r(z) = \bar{a}_x 2.08 e^{j(6z + 159.9°)}$ (V/m).

$\bar{H}_r(z) = \frac{1}{\eta_1}(-\bar{a}_z)\times\bar{E}_r(z) = \frac{1}{377}(-\bar{a}_z \times \bar{a}_x) 2.08 e^{j(6z+159.9°)}$

$= -\bar{a}_y 0.0055 e^{j(6z+159.9°)}$ (A/m).

$\bar{E}_t(z) = \bar{a}_x (\tau E_{i0}) e^{-\alpha_2 z} e^{-j\beta_2 z} = \bar{a}_x 8.08 e^{-1.35z} e^{-j(9.10z - 5.1°)}$ (V/m).

$\bar{H}_t(z) = \bar{a}_y \frac{8.08}{\eta_2} e^{-1.35z} e^{-j(9.10z - 5.1°)}$

$= \bar{a}_y 0.032 e^{-1.35z} e^{-j(9.10z + 3.4°)}$ (A/m).

b) $S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.208}{1 - 0.208} = 1.53.$

c) $(\bar{\mathscr{P}}_{av})_1 = \frac{1}{2}\mathscr{R}e\left(\bar{E}_i \times \bar{H}_i^* + \bar{E}_r \times \bar{H}_r^*\right)$

$= \bar{a}_z\left(\frac{10^2}{2\times120\pi} - \frac{2.08^2}{2\times120\pi}\right) = \bar{a}_z 0.127$ (W/m²).

$(\bar{\mathscr{P}}_{av})_2 = \frac{1}{2}\mathscr{R}e\left(\bar{E}_t \times \bar{H}_t\right) = \bar{a}_z \frac{8.08^2}{2\times254} e^{-2.70z}\cos 8.5°$

$= \bar{a}_z 0.127 e^{-2.70z}$ (W/m²).

P.7-20 Given $\bar{E}_i(x,z) = \bar{a}_y 10\, e^{-j(6x+8z)}$ (V/m).

a) $k_x = 6,\ k_z = 8 \longrightarrow k = \beta = \sqrt{k_x^2 + k_z^2} = 10$ (rad/m).

$\lambda = 2\pi/k = 2\pi/10 = 0.628$ (m); $f = c/\lambda = 4.78\times 10^8$ (Hz); $\omega = kc = 3\times 10^9$ (rad/s).

b) $\bar{E}_i(x,z;t) = \bar{a}_y 10 \cos(3\times 10^9 t - 6x - 8z)$ (V/m).

$\bar{H}_i(x,z) = \frac{1}{\eta_0}\bar{a}_{ni}\times \bar{E}_i$ $\quad \left(\bar{a}_{ni} = \frac{\bar{k}}{k} = \bar{a}_x 0.6 + \bar{a}_z 0.8.\right)$

$= \frac{1}{120\pi}(\bar{a}_x 0.6 + \bar{a}_z 0.8)\times \bar{a}_y 10\, e^{-j(6x+8z)} = \left(-\bar{a}_x \frac{1}{15\pi} + \bar{a}_z \frac{1}{20\pi}\right)e^{-j(6x+8z)}$

$\bar{H}_i(x,z;t) = \left(-\bar{a}_x \frac{1}{15\pi} + \bar{a}_z \frac{1}{20\pi}\right)\cos(3\times 10^9 t - 6x - 8z)$ (A/m).

c) $\cos\theta_i = \bar{a}_{ni}\cdot\bar{a}_z = \left(\frac{\bar{k}}{k}\right)\cdot\bar{a}_z = 0.8 \longrightarrow \theta_i = \cos^{-1}0.8 = 36.9°$.

d) $\bar{E}_i(x,0) + \bar{E}_r(x,0) = 0 \longrightarrow \bar{E}_r(x,z) = -\bar{a}_y 10\, e^{-j(6x-8z)}$.

$\bar{H}_r(x,z) = \frac{1}{\eta_0}\bar{a}_{nr}\times\bar{E}_r(x,z)$ $\quad \left(\bar{a}_{nr} = \bar{a}_x 0.6 - \bar{a}_z 0.8.\right)$

$= -\left(\bar{a}_x \frac{1}{15\pi} + \bar{a}_z \frac{1}{20\pi}\right)e^{-j(6x-8z)}$

e) $\bar{E}_1(x,z) = \bar{E}_i(x,z) + \bar{E}_r(x,z) = \bar{a}_y 10\left(e^{-j8z} - e^{j8z}\right)e^{-j6x}$

$= -\bar{a}_y j20\, e^{-j6x}\sin 8z$ (V/m).

$\bar{H}_1(x,z) = \bar{H}_i(x,z) + \bar{H}_r(x,z) = -\left(\bar{a}_x \frac{2}{15\pi}\cos 8z + \bar{a}_z \frac{j}{10\pi}\sin 8z\right)e^{-j6x}$ (A/m).

P.7-21 Snell's law of reflection: $\theta_r = \theta_i = 30°$.

Snell's law of refraction: $\sin\theta_t = \sqrt{\frac{\epsilon_1}{\epsilon_2}}\sin\theta_i = \frac{1}{3}$.

$\theta_t = 19.47°$, $\cos\theta_t = 0.943$.

$\eta_1 = \eta_0 = 377\ (\Omega)$, $\quad \eta_2 = \sqrt{\frac{\mu_2}{\epsilon_2}} = \frac{377}{\sqrt{2.25}} = 251\ (\Omega)$.

a) $\Gamma_\perp = \frac{\eta_2\cos\theta_i - \eta_1\cos\theta_t}{\eta_2\cos\theta_i + \eta_1\cos\theta_t} = -0.241$.

$\tau_\perp = 1 + \Gamma_\perp = 1 - 0.241 = 0.759$.

b) From Eq.(7-141): $\bar{E}_t(x,z) = \bar{a}_y \tau E_{i0}\, e^{-j\beta_2(x\sin\theta_t + z\cos\theta_t)}$.

$\beta_2 = \omega\sqrt{\mu_2\epsilon_2}$ $\longrightarrow \bar{E}_t(x,z;t) = \bar{a}_y 15.2\cos(2\pi 10^8 t - 1.05x - 2.96z)$ (V/m).

$= \pi$ (rad/m).

From Eq.(7-142): $\bar{H}_t(x,z) = \frac{15.2}{251}\left(-\bar{a}_x\cos\theta_t + \bar{a}_z\sin\theta_t\right)e^{-j(1.05x+2.96z)}$

$\longrightarrow \bar{H}_t(x,z;t) = 0.06(-\bar{a}_x 0.943 + \bar{a}_z 0.333)\cos(2\pi 10^8 t - 1.05x - 2.96z)$ (A/m).

P.7-22 From problem P.7-21:

$$\theta_r = \theta_i = 30^\circ, \qquad \theta_t = 19.47^\circ.$$

$$\eta_1 = 377\ (\Omega), \quad \eta_2 = 251\ (\Omega).$$

a) From Eq. (7-158): $\Gamma_{\parallel} = \dfrac{\eta_2 \cos\theta_t - \eta_1 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$.

$$\Gamma_{\parallel} = \frac{251\times 0.943 - 377\times 0.866}{251\times 0.943 + 377\times 0.866} = -0.159.$$

From Eq. (7-160):

$$\tau_{\parallel} = \left(1 + \Gamma_{\parallel}\right)\frac{\cos\theta_i}{\cos\theta_t} = 0.772.$$

b) $\bar{H}_i(x,z) = \bar{a}_y 0.053\, e^{-j\beta_1(x\sin\theta_i + z\cos\theta_i)}$.

$$\beta_1 = \frac{\omega}{c} = \frac{2\pi\times 10^8}{3\times 10^8} = \frac{2\pi}{3} \quad (\text{rad/m}).$$

$$\beta_2 = \omega\sqrt{\mu_2\epsilon_2} = \sqrt{\epsilon_{r2}}\,\beta_1 = \pi \quad (\text{rad/m}).$$

$$\bar{H}_i(x,z) = \bar{a}_y 0.053\, e^{-j(\pi x/3 + \pi z/\sqrt{3})} \quad (\text{A/m}).$$

From Eq. (7-150): $\bar{E}_i(x,z) = 19.98(\bar{a}_x 0.866 - \bar{a}_z 0.5)e^{-j(\pi x/3 + \pi z/\sqrt{3})}$ (V/m).

$$E_{t0} = \tau_{\parallel} E_{i0} = 0.772\times 19.98 = 15.42\ (\text{V/m}).$$

From Eqs. (7-154) and (7-155):

$$\bar{E}_t(x,z) = 15.42(\bar{a}_x 0.943 - \bar{a}_z 0.333)e^{-j(1.05x + 2.96z)}.$$

$$\bar{H}_t(x,z) = \bar{a}_y 0.061\, e^{-j(1.05x + 2.96z)}.$$

Thus, with a cosine reference,

$$\bar{E}_t(x,z;t) = 15.42(\bar{a}_x 0.943 - \bar{a}_z 0.333)\cos(2\pi 10^8 t - 1.05x - 2.96z)\ (\text{V/m}).$$

$$\bar{H}_t(x,z;t) = \bar{a}_y 0.061\cos(2\pi 10^8 t - 1.05x - 2.96z)\ (\text{A/m}).$$

P.7-24 Given $f = f_p/2$ and $\theta_i = 60°$.

$$\longrightarrow \eta_p = \eta_0 / \sqrt{1-(f_p/f)^2} = -j\eta_0/\sqrt{3}, \quad \eta_p/\eta_0 = -j/\sqrt{3}.$$

From Eq.(7-118): $\sin\theta_t = \frac{\eta_p}{\eta_0}\sin\theta_i = -j/2$, $\cos\theta_t = \sqrt{5}/2$, $\cos\theta_i = 1/2$.

a) From Eq. (7-147): $\Gamma_\perp = \frac{(\eta_p/\eta_0)\cos\theta_i - \cos\theta_t}{(\eta_p/\eta_0)\cos\theta_i + \cos\theta_t} = e^{j209°}$,

From Eq. (7-148): $\tau_\perp = \frac{2(\eta_p/\eta_0)\cos\theta_i}{(\eta_p/\eta_0)\cos\theta_i + \cos\theta_t} = 0.5\, e^{-j75.5°}$,

b) From Eq. (7-158): $\Gamma_\parallel = \frac{(\eta_p/\eta_0)\cos\theta_t - \cos\theta_i}{(\eta_p/\eta_0)\cos\theta_t + \cos\theta_i} = e^{j76°}$,

From Eq. (7-159): $\tau_\parallel = \frac{2(\eta_p/\eta_0)\cos\theta_i}{(\eta_p/\eta_0)\cos\theta_t + \cos\theta_i} = 0.177\, e^{-j38°}$.

$|\Gamma_\perp| = |\Gamma_\parallel| = 1$, but the phase shift of the reflected wave depends on the polarization of the incident wave. There are standing waves in the air and exponentially decaying transmitted waves in the ionosphere.

P.7-25 $k_{2x}^2 + k_{2z}^2 = k_2^2 = \omega^2\mu_0\epsilon_2 - j\omega\mu_0\sigma_2$. ①

Continuity conditions at $z=0$ for all x and y require:

$$k_{2x} = k_{1x} = \omega\sqrt{\mu_0\epsilon_0}\sin\theta_i = \beta_x = 2.09\times10^{-4}. \quad ②$$

$$k_{2z} = \beta_{2z} - j\alpha_{2z}. \quad ③$$

Combining ①, ② and ③, we can solve for α_{2z} and β_{2z} in terms of ω, μ_0, ϵ_2, σ_2, and β_x. But, since

$$\beta_x^2 \ll \omega^2\mu_0\epsilon_2,$$

we have $\alpha_2 = \alpha_{2z} \cong \beta_{2z} \cong \frac{1}{\delta} = \sqrt{\pi f\mu_0\sigma_2} = 0.3974$ (m^{-1}).

a) $\theta_t = \tan^{-1}\frac{\beta_x}{\beta_{2z}} \cong \tan^{-1}\frac{2.09}{0.3974}\times10^{-4} \cong 5.26\times10^{-4}$ (rad)
$= 0.03°$.

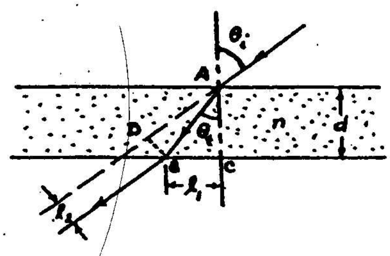
b) $\Gamma_{\parallel} = \dfrac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_0 \cos\theta_i}$ $\qquad \eta_2 = \dfrac{\alpha_2}{\sigma_2}(1+j) = 0.0993(1+j)$.

$= \dfrac{2\times 0.0993(1+j)}{0.0993(1+j) + 377\cos 88^\circ}$ $\qquad \cos\theta_t = \cos 0.03^\circ \cong 1$.

$\cong 0.0151(1+j) = 0.0214\, e^{j\pi/4}$.

c) $20\log_{10} e^{-\alpha_2 z} = -30. \longrightarrow z = \dfrac{1.5}{\alpha_2 \log_{10} e} = 8.69$ (m).

P.7-26



a) Snell's law:

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{1}{n},$$

$$\theta_t = \sin^{-1}\left(\frac{1}{n}\sin\theta_i\right).$$

b) $\cos\theta_t = \sqrt{1-\left(\frac{1}{n}\sin\theta_i\right)^2}$.

$$\ell_1 = \overline{BC} = \overline{AC}\tan\theta_t = d\,\frac{\sin\theta_t}{\cos\theta_t} = \frac{d\sin\theta_i}{\sqrt{n^2-\sin^2\theta_i}}.$$

c) $\ell_2 = \overline{BD} = \overline{AC}\,\sin(\theta_i-\theta_t) = \dfrac{d}{\cos\theta_t}(\sin\theta_i\cos\theta_t - \cos\theta_i\sin\theta_t)$

$$= d\sin\theta_i\left[1-\frac{\cos\theta_i}{\sqrt{n^2-\sin^2\theta_i}}\right].$$

P.7-27

a) $\sin\theta_c = \sqrt{\dfrac{\epsilon_2}{\epsilon_1}}. \longrightarrow \sin\theta_t = \sqrt{\dfrac{\epsilon_1}{\epsilon_2}}\sin\theta_i > 1$ for $\theta_i > \theta_c$.

$$\cos\theta_t = -j\sqrt{\left(\frac{\epsilon_1}{\epsilon_2}\right)\sin^2\theta_i - 1}.$$

From Eqs. (7-141) and (7-142):

$$\bar{E}_t(x,z) = \bar{a}_y E_{to}\, e^{-\alpha_2 z} e^{-j\beta_{2x}x},$$

$$\bar{H}_t(x,z) = \frac{E_{to}}{\eta_2}\left(\bar{a}_x\, j\alpha_2 + \bar{a}_z\sqrt{\frac{\epsilon_1}{\epsilon_2}}\sin\theta_i\right)e^{-\alpha_2 z}e^{-j\beta_{2x}x},$$

where $\beta_{2x} = \beta_2\sin\theta_t = \beta_2\sqrt{\dfrac{\epsilon_1}{\epsilon_2}}\sin\theta_i$,

$$\alpha_2 = \beta_2\sqrt{\left(\frac{\epsilon_1}{\epsilon_2}\right)\sin^2\theta_i - 1},$$

$$E_{to} = \frac{2\eta_2\cos\theta_i\, E_{io}}{\eta_2\cos\theta_i - j\eta_1\sqrt{\left(\frac{\epsilon_1}{\epsilon_2}\right)\sin^2\theta_i - 1}} \quad \text{from Eq. (7-148).}$$

b) $(\mathcal{P}_{av})_{2z} = \dfrac{1}{2}\mathcal{R}e\,(E_{ty}H_{tx}^{*}) = 0.$

P. 7-28 Given $\theta_i = \theta_c$. $\longrightarrow$ $\theta_t = \pi/2$, $\cos\theta_t = 0$.

a) From Eq. (7-148): $(E_{t0}/E_{i0})_\perp = 2$.

b) From Eq. (7-159): $(E_{t0}/E_{i0})_\parallel = 2\eta_2/\eta_1$.

c) $\bar{E}_i(x,z;t) = \bar{a}_y E_{i0} \cos\omega\left[t - \frac{n_1}{c}(x\sin\theta_i + z\cos\theta_i)\right]$,

$$\bar{E}_t(x,z;t) = \bar{a}_y 2E_{i0} e^{-\alpha z}\cos\omega(t - \tfrac{n_2}{c}x\sin\theta_t)$$
$$= \bar{a}_y 2E_{i0} e^{-\alpha z}\cos\omega(t - \tfrac{n_1}{c}x\sin\theta_i),$$

where $\alpha = \frac{n_2\omega}{c}\sqrt{(\frac{n_1}{n_2}\sin\theta_i)^2 - 1} = 0$, when $\theta = \theta_c$.

P. 7-29 a) $\theta_c = \sin^{-1}\sqrt{\epsilon_{r2}/\epsilon_{r1}} = \sin^{-1}\sqrt{1/81} = 6.38°$.

b) $\theta_i = 20° > \theta_c$. $\sin\theta_t = \sqrt{\frac{\epsilon_1}{\epsilon_2}}\sin\theta_i = 3.08$. $\cos\theta_t = -j2.91$.

$$\Gamma_\perp = \frac{\sqrt{\epsilon_{r1}}\cos\theta_i - \cos\theta_t}{\sqrt{\epsilon_{r1}}\cos\theta_i + \cos\theta_t} = e^{j38°} = e^{j0.66}$$

c) $$\tau_\perp = \frac{2\sqrt{\epsilon_{r1}}\cos\theta_i}{\sqrt{\epsilon_{r1}}\cos\theta_i + \cos\theta_t} = 1.89\, e^{j19°} = 1.89\, e^{j0.33}$$

d) The transmitted wave in air varies as $e^{-\alpha_2 z}e^{-j\beta_{2x}x}$.
where $\alpha_2 = \beta_2\sqrt{(\frac{\epsilon_1}{\epsilon_2})\sin^2\theta_i - 1} = \frac{2\pi}{\lambda_0}(2.91)$.

Attenuation in air for each wavelength
$= 20\log_{10} e^{-\alpha_2\lambda_0} = 159$ (dB).

P. 7-30 When the incident light first strikes the hypotenuse surface, $\theta_i = \theta_t = 0$, $\tau_1 = \frac{2\eta_2}{\eta_2 + \eta_0}$.

$$\frac{(\mathcal{P}_{av})_{t1}}{(\mathcal{P}_{av})_i} = \frac{\eta_0}{\eta_2}\tau_1^2 = \frac{4\eta_0\eta_2}{(\eta_2 + \eta_0)^2}.$$

Total reflections occur inside the prism at both slanting surfaces because

$$\theta_i = 45° > \theta_c = \sin^{-1}(\tfrac{1}{2}) = 30°.$$

On exit from the prism, $\tau_2 = \dfrac{2\eta_0}{\eta_2+\eta_0}$.

$$\frac{(\mathcal{P}_{av})_o}{(\mathcal{P}_{av})_{t1}} = \frac{\eta_2}{\eta_0}\tau_2^2 = \frac{4\eta_0\eta_2}{(\eta_2+\eta_0)^2}.$$

$$\therefore \quad \frac{(\mathcal{P}_{av})_o}{(\mathcal{P}_{av})_i} = \left[\frac{4\eta_0\eta_2}{(\eta_2+\eta_0)^2}\right]^2 = \left[\frac{4\sqrt{\epsilon_r}}{(1+\sqrt{\epsilon_r})^2}\right]^2 = 0.79.$$

<u>P.7-31</u> a) $n_0 \sin\theta_a = n_1 \sin(90° - \theta_c) = n_1 \cos\theta_c$

$$= n_1\sqrt{1-\sin^2\theta_c} = n_1\sqrt{1-(n_2/n_1)^2} = \sqrt{n_1^2-n_2^2}.$$

$$\sin\theta_a = \frac{1}{n_0}\sqrt{n_1^2-n_2^2} = \sqrt{n_1^2-n_2^2}. \qquad (n_0=1)$$

b) $N.A. = \sin\theta_a = \sqrt{2^2-1.74^2} = 0.9861,$

$$\theta_a = \sin^{-1}0.9861 = 80.4°.$$

<u>P.7-32</u>

a) For perpendicular polarization and $\mu_1 \neq \mu_2$:

$$\sin\theta_{B\perp} = \frac{1}{\sqrt{1+\left(\frac{\mu_1}{\mu_2}\right)}}.$$

Under condition of no reflection:

$$\cos\theta_t = \sqrt{1-\frac{n_1^2}{n_2^2}\sin^2\theta_{B\perp}} = \frac{1}{\sqrt{1+\left(\frac{\mu_2}{\mu_1}\right)}}$$

$$= \sin\theta_{B\perp} \longrightarrow \theta_t + \theta_{B\perp} = \pi/2.$$

b) For parallel polarization and $\epsilon_1 \neq \epsilon_2$:

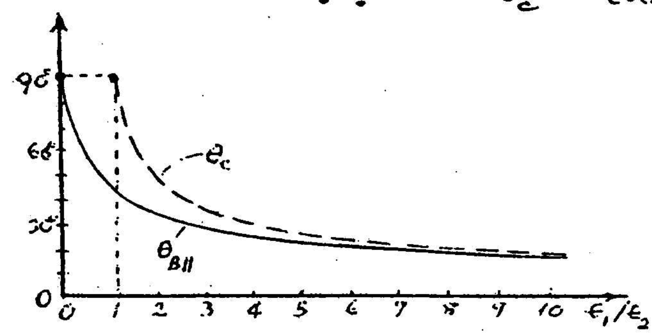
$$\sin\theta_{B\parallel} = \frac{1}{\sqrt{1+\left(\frac{\epsilon_1}{\epsilon_2}\right)}}.$$

$$\cos\theta_t = \sqrt{1-\frac{n_1^2}{n_2^2}\sin^2\theta_{B\parallel}} = \frac{1}{\sqrt{1+\left(\frac{\epsilon_2}{\epsilon_1}\right)}}$$

$$= \sin\theta_{B\parallel} \longrightarrow \theta_t + \theta_{B\parallel} = \pi/2.$$

P.7-33 For two contiguous media with equal permeability and permittivities ϵ_1 and ϵ_2, we have from Eq. (7-120): $\theta_c = \sin^{-1}\sqrt{\epsilon_2/\epsilon_1}$, and from Eq. (7-164): $\theta_{B\parallel} = \tan^{-1}\sqrt{\epsilon_2/\epsilon_1}$.

$$\therefore \quad \sin\theta_c = \tan\theta_{B\parallel}.$$



ϵ_1/ϵ_2	θ_c	$\theta_{B\parallel}$
0	—	90°
0.5	—	54.7°
1	90°	45°
2	45°	35.3°
4	30°	26.6°
8	20.7°	19.5°
10	18.4°	17.6°

Chapter 8

Transmission Lines

P. 8-1 Substituting Eqs. (8-17) and (8-18) in Eq. (8-43):

$$Z_0 = \frac{d}{w}\sqrt{\frac{\mu}{\epsilon}}$$

a) $Z_0 = \frac{d'}{w}\sqrt{\frac{\mu}{2\epsilon}} = \frac{d}{w}\sqrt{\frac{\mu}{\epsilon}} \longrightarrow d' = \sqrt{2}\, d.$

b) $Z_0 = \frac{d}{w'}\sqrt{\frac{\mu}{2\epsilon}} = \frac{d}{w}\sqrt{\frac{\mu}{\epsilon}} \longrightarrow w' = \frac{1}{\sqrt{2}} w.$

c) $Z_0 = \frac{2d}{w'}\sqrt{\frac{\mu}{\epsilon}} = \frac{d}{w}\sqrt{\frac{\mu}{\epsilon}} \longrightarrow w' = 2w.$

d) $u_p = \frac{1}{\sqrt{\mu\epsilon}} \longrightarrow u_{pa} = u_p/\sqrt{2}$ for part a.

$u_{pb} = u_p/\sqrt{2}$ for part b.

$u_{pc} = u_p$ for part c.

P. 8-2 Given: $\sigma_c = 1.6\times10^7$ (S/m), $w = 0.02$ (m), $d = 2.5\times10^{-3}$ (m)

Lossy dielectric slab: $\mu = \mu_0$, $\epsilon_r = 3$, $\sigma = 10^{-3}$ (S/m).

$f = 5\times10^8$ (Hz).

a) $R = \frac{2}{w}\sqrt{\frac{\pi f \mu_0}{\sigma_c}} = 1.11$ (Ω/m).

$L = \mu\frac{d}{w} = 0.157$ (μH/m).

$G = \sigma\frac{w}{d} = 0.008$ (S/m).

$C = \epsilon\frac{w}{d} = 0.212$ (nF/m).

b) $\frac{|E_z|}{|E_y|} = \sqrt{\frac{\omega\epsilon}{\sigma_c}} = 4.167\times10^{-5}.$

c) $\omega L = 493.5 \gg R$, $\omega C = 0.667 \gg G$.

$\gamma \cong j\omega\sqrt{LC}\left[1+\frac{1}{2j}\left(\frac{R}{\omega L}+\frac{G}{\omega C}\right)\right] = 0.129 + j18.14$ (m^{-1}),

$Z_0 \cong \sqrt{\frac{L}{C}}\left[1+\frac{1}{2j}\left(\frac{R}{\omega L}-\frac{G}{\omega C}\right)\right] = 27.21 + j0.13$ (Ω).

P. 8-3 a) For two-wire transmission line:

$$Z_0 = \sqrt{\frac{L}{C}} = \frac{1}{\pi}\sqrt{\frac{\mu}{\epsilon}}\cosh^{-1}\left(\frac{D}{2a}\right) = \frac{120}{\sqrt{\epsilon_r}}\ln\left[\frac{D}{2a}+\sqrt{\left(\frac{D}{2a}\right)^2-1}\right] = 300\ (\Omega).$$

$$\frac{D}{2a} = 21.27. \longrightarrow D = 25.5\times10^{-3}\ (m).$$

b) For coaxial transmission line:

$$Z_0 = \frac{1}{2\pi}\sqrt{\frac{\mu}{\epsilon}}\ln\left(\frac{b}{a}\right) = \frac{60}{\sqrt{\epsilon_r}}\ln\left(\frac{b}{a}\right) = 75.$$

$$\frac{b}{a} = 6.52 \longrightarrow b = 3.91\times10^{-3}\ (m).$$

P. 8-4 From Eq. (8-61): $\alpha = R\sqrt{\frac{C}{L}}$.

From Eqs. (8-28) and (8-29): $\alpha = \dfrac{R}{\frac{1}{2\pi}\sqrt{\frac{\mu}{\epsilon}}\ln\frac{b}{a}} = \dfrac{R}{75}$.

From Table 8-1: $R = \dfrac{R_s}{2\pi}\left(\dfrac{1}{a}+\dfrac{1}{b}\right)$.

For copper at 1 (MHz): $R_s = \sqrt{\dfrac{\pi f\mu_c}{\sigma_c}} = \sqrt{\dfrac{\pi 10^6\times4\times10^{-7}}{5.8\times10^7}}$

$$= 2.61\times10^{-4}\ (\Omega).$$

$$R = \frac{2.61\times10^{-4}}{2\pi}\left(\frac{1}{0.6}+\frac{1}{3.91}\right)\times10^3 = 0.08\ (\Omega).$$

$$\therefore\ \alpha = \frac{0.08}{75} = 1.065\times10^{-3}\ (Np/m)$$
$$= 9.25\times10^{-3}\ (dB/m).$$

P. 8-5 Eq. (8-38): $Z_0 = \sqrt{\dfrac{R+j\omega L}{G+j\omega C}}$.

Given $Z_0 = 50+j0\ (\Omega)$ — Purely real.

$Im(Z_0) = 0 \longrightarrow \dfrac{R}{L} = \dfrac{G}{C} = k$ (Distortionless line).

Given: $\alpha = 0.01\ (dB/m) = 0.00115\ (Np/m)$.
$\beta = 0.8\pi\ (rad/m)$; $f = 10^8\ (Hz)$.

From Eqs. (8-48), (8-49) and (8-51): $\alpha = R\sqrt{\frac{C}{L}}$, $\beta = \omega\sqrt{LC}$, $Z_0 = \sqrt{\frac{L}{C}}$.

$R = \alpha Z_0 = 0.0576\ (\Omega/m)$, $L = \dfrac{\beta Z_0}{2\pi f} = 0.20\ (\mu H/m)$,

$G = \dfrac{RC}{L} = \dfrac{\alpha}{Z_0} = 23\ (\mu S/m)$, $C = \dfrac{L}{Z_0^2} = 80\ (pF/m)$.

P. 8-6 $(P_{av})_L = (P_{av})_i = \frac{1}{2}\mathcal{R}e[V_i I_i^*]$ $\qquad V_i = \frac{Z_i}{Z_g+Z_i}V_g$,

$$= \frac{|V_g|^2 R_i}{(R_g+R_i)^2+(X_g+X_i)^2}. \qquad I_i = \frac{V_g}{Z_g+Z_i}.$$

To maximize $(P_{av})_L$, set $\left.\begin{array}{l}\dfrac{\partial (P_{av})_L}{\partial R_i} = 0. \\ \text{and } \dfrac{\partial (P_{av})_L}{\partial X_i} = 0.\end{array}\right\} R_i = R_g,\ X_i = -X_g$ or $Z_i = Z_g^*$.

$$\text{Max.}\ (P_{av})_L = \frac{|V_g|^2}{4R_g} = (P_{av})_{Z_g}.$$

$\longrightarrow$ Max. power-transfer efficiency = 50%.

P. 8-7 $V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$,

$I(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}$.

At $z=0$; $V(0) = V_i = V_0^+ + V_0^-$, $\quad I(0) = I_i = I_0^+ + I_0^- = \frac{1}{Z_0}(V_0^+ - V_0^-)$.

$\longrightarrow V_0^+ = \frac{1}{2}(V_i + I_i Z_0)$, $\quad V_0^- = \frac{1}{2}(V_i - I_i Z_0)$.

a) $V(z) = \frac{1}{2}(V_i + I_i Z_0)e^{-\gamma z} + \frac{1}{2}(V_i - I_i Z_0)e^{\gamma z}$,

$I(z) = \frac{1}{2Z_0}(V_i + I_i Z_0)e^{-\gamma z} - \frac{1}{2Z_0}(V_i - I_i Z_0)e^{\gamma z}$.

b) $V(z) = V_i \cosh \gamma z - I_i Z_0 \sinh \gamma z$,

$I(z) = I_i \cosh \gamma z - \frac{V_i}{Z_0}\sinh \gamma z$.

P. 8-8 a) $-\frac{dV}{dz} = RI$, $\quad -\frac{dI}{dz} = GV$.

$$\begin{cases} \dfrac{d^2V}{dz^2} = RGV, \\ \dfrac{d^2I}{dz^2} = RGI. \end{cases}$$

b) $V(z) = V_0^+ e^{-\alpha z} + V_0^- e^{\alpha z}$,

$I(z) = I_0^+ e^{-\alpha z} + I_0^- e^{\alpha z}$, $\quad \alpha = \sqrt{RG}$.

$$\frac{V_0^+}{I_0^+} = -\frac{V_0^-}{I_0^-} = R_0 = \sqrt{\frac{R}{G}}.$$

We have $V(z) = \frac{1}{2}(V_i + I_i R_0)e^{-\alpha z} + \frac{1}{2}(V_i - I_i R_0)e^{\alpha z},$

$$I(z) = \frac{1}{2}\left(\frac{V_i}{R_0} + I_i\right)e^{-\alpha z} - \frac{1}{2}\left(\frac{V_i}{R_0} - I_i\right)e^{\alpha z},$$

where $V_i = \frac{R_i}{R_g + R_i} V_g$ and $I_i = \frac{V_g}{R_g + R_i}$.

c) For an infinite line, $R_i = R_0$:

$$V(z) = \frac{R_0}{R_g + R_0} V_g e^{-\alpha z}, \qquad I(z) = \frac{V_g}{R_g + R_0} e^{-\alpha z}.$$

d) For a finite line of length ℓ terminated in R_L:

$$R_i = R_0 \frac{R_L + R_0 \tanh \alpha \ell}{R_0 + R_L \tanh \alpha \ell}.$$

P.8-9 Distortionless line: $R_0 = \sqrt{\frac{L}{C}} = 50\ (\Omega)$, $R = 0.5\ (\Omega/m)$,

$$\tan\left(\frac{\sigma_d}{\omega \epsilon}\right) = \tan\left(\frac{G}{\omega C}\right) = 0.0018.$$

$$\longrightarrow \frac{G}{\omega C} \cong 0.0018; \quad \frac{G}{C} = 8000\pi \times 0.0018 = 45.2 = \frac{R}{L}.$$

$$L = \frac{R}{G/C} = 0.011\ (H/m), \quad C = \frac{L}{R_0^2} = 4.42\ (\mu F/m).$$

$$\alpha = \frac{R}{R_0} = 0.010\ (Np/m), \quad \beta = \omega\sqrt{LC} = 5.55\ (rad/m).$$

a) $V(z) = \frac{V_{g0} R_0}{R_0 + R_g} e^{-\alpha z} e^{-j\beta z} = \frac{50}{9 + j3} e^{-0.01z} e^{-j5.55z}; \quad I(z) = \frac{V(z)}{50}.$

$\therefore V(z,t) = 5.27\, e^{-0.01z} \sin(8000\pi t - 5.55z - 0.322) \quad (V),$

$I(z,t) = 0.105\, e^{-0.01z} \sin(8000\pi t - 5.55z - 0.322) \quad (A).$

b) At $z = 50\,(m)$: $V(50,t) = 3.20 \sin(8000\pi t - 0.432\pi) \quad (V),$

$I(50,t) = 0.064 \sin(8000\pi t - 0.432\pi) \quad (A).$

c) $(P_{av})_L = \frac{1}{2}\mathcal{R}e|V_L I_L^*| = \frac{1}{2}(3.20 \times 0.064) = 0.102\ (W).$

P.8-10 a) For a short-circuited line, set $Z_L=0$ in Eq.(8-78) to obtain:

$$Z_{is}=Z_0\tanh\gamma\ell=Z_0\frac{1-e^{-2\gamma\ell}}{1+e^{-2\gamma\ell}}.$$

For $\ell=\lambda/4$, $\beta\ell=\pi/2$, $\alpha\lambda/2\ll 1$.

$$Z_{is}=Z_0\frac{1-e^{-2\alpha(\lambda/4)}e^{-j\pi}}{1+e^{-2\alpha(\lambda/4)}e^{-j\pi}}\cong Z_0\frac{1+(1-\alpha\lambda/2)}{1-(1-\alpha\lambda/2)}$$

$$\cong 4Z_0/\alpha\lambda.$$

b) For an open-circuited line, set $Z_L\to\infty$ in Eq.(8-78) to obtain:

$$Z_{io}=Z_0\coth\gamma\ell=Z_0\frac{1+e^{-2\gamma\ell}}{1-e^{-2\gamma\ell}}.$$

For $\ell=\lambda/4$, $Z_{io}=Z_0\frac{1+e^{-(\alpha\lambda/2)}e^{-j\pi}}{1-e^{-(\alpha\lambda/2)}e^{-j\pi}}\cong Z_0\frac{1-(1-\alpha\lambda/2)}{1+(1-\alpha\lambda/2)}$

$$\cong Z_0\alpha\lambda/4.$$

P.8-11 $\beta\ell=\frac{2\pi f}{c}\ell=\frac{8\pi}{3}=480°.$

$\tan\beta\ell=\tan 480°=-1.732,$

$$Z_i=Z_0\frac{Z_L+jZ_0\tan\beta\ell}{Z_0+jZ_L\tan\beta\ell}=50\frac{(40+j30)+j50(-1.732)}{50+j(40+j30)(-1.732)}$$

$$=26.3-j9.87\quad(\Omega).$$

P.8-12 Given: $Z_{io}=Z_0\coth\gamma\ell=250\angle{-50°}\quad(\Omega),$

$Z_{is}=Z_0\tan\gamma\ell=360\angle{20°}\quad(\Omega).$

a) $Z_0=\sqrt{Z_{io}Z_{is}}=300\angle{-15°}=289.8-j77.6\quad(\Omega).$

$\tanh\gamma\ell=\sqrt{\frac{Z_{is}}{Z_{io}}}=1.2\angle{35°}=0.983+j.688=\tanh(\alpha\ell+j\beta\ell).$

$\ell=4\ (m)\longrightarrow\quad \alpha=0.139\quad(Np/m),$

$\beta=0.235\quad(rad/m).$

b) $Z_0=\sqrt{\frac{R+j\omega L}{G+j\omega C}},\quad \gamma=\sqrt{(R+j\omega L)(G+j\omega C)}.$

$\longrightarrow\quad R+j\omega L=\gamma Z_0;\quad G+j\omega C=\frac{\gamma}{Z_0}.$

$\omega=\beta c=0.235\times3\times10^8=0.705\times10^8\ (rad/m).$

We obtain: $R=58.6\ (\Omega),\qquad L=0.812\ (\mu H/m),$

$G=0.246\ (mS/m),\quad C=12.4\quad(pF/m).$

P. 8-13 a) Since the line is very short compared to a wavelength, we may use Eqs. (8-81) and (8-83).

$$\therefore \quad \left.\begin{aligned} C &= \frac{54\times10^{-12}}{0.6} = 9\times10^{-11}\ (F/m), \\ L &= \frac{0.3\times10^{-6}}{0.6} = 5\times10^{-7}\ (H/m). \end{aligned}\right\} R_0 = \sqrt{\frac{L}{C}} = 74.5\ (\Omega).$$

$$\mu\epsilon = LC \longrightarrow \epsilon_r = \frac{LC}{\mu_0\epsilon_0} = 4.05.$$

b) $$\beta = \frac{\omega}{u_p} = 2\pi\times10^7\sqrt{LC} = 0.42\ (rad/m); \quad \beta\ell = 0.42\times0.6 = 0.252\ (rad) = 14.4^\circ$$

$$\therefore X_{io} = -R_0 \cot\beta\ell \doteq -\frac{1}{\omega C\ell} = -290\ (\Omega),$$

$$X_{is} = R_0 \tan\beta\ell \doteq \omega L\ell = 19.2\ (\Omega).$$

P. 8-14 For load impedance

$$Z_L = R_L + jX_L,$$

a) $$|\Gamma| = \frac{S-1}{S+1} = \frac{\left|\frac{Z_L}{Z_0} - 1\right|}{\left|\frac{Z_L}{Z_0} + 1\right|} = \frac{\sqrt{(r_L-1)^2 + x_L^2}}{\sqrt{(r_L+1)^2 + x_L^2}},$$

where $r_L = R_L/Z_0$ and $x_L = X_L/Z_0$.

$$\longrightarrow x_L = \pm\left[\frac{\left(\frac{S-1}{S+1}\right)^2 (r_L+1)^2 - (r_L-1)^2}{1 - \left(\frac{S-1}{S+1}\right)^2}\right].$$

When $S = 3$, $x_L = \pm\sqrt{(10r_L - 3r_L^2 - 3)/3}$.

b) $S = 3$ and $r_L = 150/75 = 2$.

$$\longrightarrow x_L = \pm\sqrt{5/3}.$$

$$X_L = x_L Z_0 = \pm 96.8\ (\Omega).$$

P.8-15 For a lossless line, $Z_0 = R_0$.

$$|\Gamma|^2 = \left|\frac{(R_L - R_0) + jX_L}{(R_L + R_0) + jX_L}\right|^2 = \frac{(R_L - R_0)^2 + X_L^2}{(R_L + R_0)^2 + X_L^2}.$$

a) Set $\frac{\partial|\Gamma|^2}{\partial R_0} = 0. \longrightarrow R_0 = \sqrt{R_L^2 + X_L^2}.$

(A minimum S corresponds to a minimum $|\Gamma|$.)

For $Z_L = 40 + j30\,(\Omega)$, $R_0 = \sqrt{40^2 + 30^2} = 50\,(\Omega)$.

b) $\text{Min.}\,|\Gamma| = \sqrt{\frac{R_0 - R_L}{R_0 + R_L}} = \sqrt{\frac{50-40}{50+40}} = \frac{1}{3}.$

$\text{Min.}\, S = \frac{1 + \frac{1}{3}}{1 - \frac{1}{3}} = 2.$

P.8-16 Lossless line of characteristic resistance R_0', length ℓ' and terminating in Z_L: (from Eq.8-79)

$$Z_i = R_0' \frac{Z_L + jR_0' t}{R_0' + jZ_L t}, \quad t = \tan\beta\ell'.$$

$$\longrightarrow Z_L = R_0' \frac{Z_i - jR_0' t}{R_0' - jZ_i t}.$$

Now set $Z_i = 50\,(\Omega)$ and $Z_L = 40 + j10\,(\Omega)$.

$$40 + j10 = R_0' \frac{50 - jR_0' t}{R_0' - j50t}.$$

$$\longrightarrow \begin{cases} 40R_0' + 500t = 50R_0', \\ 10R_0' - 2000t = -(R_0')^2 t. \end{cases}$$

Solving: $R_0' = 38.7\,(\Omega)$,

and $t = \tan\beta\ell' = 0.775.$

$\longrightarrow \ell' = 0.105\lambda.$

P. 8-17 Eq. (8-79): $R_i + jX_i = R_0 \dfrac{R_L + jR_0 \tan\beta\ell}{R_0 + jR_L \tan\beta\ell}$.

Let $r_i = \dfrac{R_i}{R_0}$, $x_i = \dfrac{X_i}{R_0}$, $r_L = \dfrac{R_L}{R_0}$, and $t = \tan\beta\ell$.

$$r_i + jx_i = \frac{r_L + jt}{1 + jr_L t}.$$

$$\longrightarrow \begin{cases} r_L(1 + x_i t) = r_i, \\ t(1 - r_L r_i) = x_i. \end{cases}$$

Solving, we obtain:

$$r_L = \frac{1}{2r_i}\left\{(1 + r_i^2 + x_i^2) \pm \sqrt{(1 + r_i^2 + x_i^2)^2 - 4r_i^2}\right\},$$

$$t = \frac{1}{2x_i}\left\{-\left[1 - (r_i^2 + x_i^2)\right] \pm \sqrt{\left[1 - (r_i^2 + x_i^2)\right]^2 + 4x_i^2}\right\},$$

$$\ell = \frac{\lambda}{2\pi}\tan^{-1} t.$$

P. 8-18 a) $|\Gamma| = \dfrac{S-1}{S+1} = \dfrac{2-1}{2+1} = \dfrac{1}{3}$.

To find θ_Γ, write Eq. (8-72) as

$$V(z') = \frac{I_L}{2}(Z_L + R_0)e^{j\beta z'}[1 + \Gamma e^{-j2\beta z'}]$$
$$= \frac{I_L}{2}(Z_L + R_0)e^{j\beta z'}[1 + |\Gamma| e^{j\theta_\Gamma} e^{-j2\beta z'}]$$
$$= \frac{I_L}{2}(Z_L + R_0)e^{j\beta z'}[1 + |\Gamma| e^{j\phi}], \quad \phi = \theta_\Gamma - 2\beta z'.$$

Voltage is minimum when $\phi = \pm\pi$,

or when $\theta_\Gamma = 2\left(\frac{2\pi}{\lambda}\right) \times 0.3\lambda - \pi = 0.2\pi$.

$$\therefore \quad \Gamma = \frac{1}{3}e^{j0.2\pi} = 0.270 + j0.196.$$

b) $Z_L = R_0 \left(\dfrac{1+\Gamma}{1-\Gamma}\right) = 300\left(\dfrac{1.270 + j0.196}{0.730 - j0.196}\right)$

$= 466 + j206 \ (\Omega)$.

P. 8-19 Given: $V_g = 0.1\angle 0°$ (V), $Z_g = Z_0 = 50\ (\Omega)$,

$R_L = 25\ (\Omega) = 0.5 Z_0$. $\ell = \frac{\lambda}{8}$.

$$\longrightarrow \Gamma_L = \frac{R_L - R_0}{R_L + R_0} = -\frac{1}{3}.$$

a)

From Fig. 8-5,

$$V_i = \frac{Z_i}{Z_0 + Z_i} V_g, \qquad I_i = \frac{V_g}{Z_0 + Z_i},$$

where from Eq. (8-78),

$$Z_i = Z_0 \frac{0.5 Z_0 + jZ_0 \tan\beta\ell}{Z_0 + j0.5 Z_0 \tan\beta\ell} = Z_0 \frac{1 + j2\tan\beta\ell}{2 + j\tan\beta\ell}.$$

$$\therefore\ V_i = \frac{1 + j2\tan\beta\ell}{3(1 + j\tan\beta\ell)} V_g = \frac{1}{30}\left(\frac{1 + j2\tan\beta\ell}{1 + j\tan\beta\ell}\right) \quad (V).$$

$$I_i = \frac{2 + j\tan\beta\ell}{3Z_0(1 + j\tan\beta\ell)} V_g = \frac{2}{3}\left(\frac{2 + j\tan\beta\ell}{1 + j\tan\beta\ell}\right) \quad (mA).$$

For $\ell = \frac{\lambda}{8}$, $\beta\ell = \left(\frac{2\pi}{\lambda}\right)\frac{\lambda}{8} = \frac{\pi}{4}$, $\tan\beta\ell = 1$.

$$V_i = \frac{1}{30}\left(\frac{1+j2}{1+j1}\right) = 0.527\angle{+18.4°} \quad (V).$$

$$I_i = \frac{2}{3}\left(\frac{2+j1}{1+j1}\right) = 1.054\angle{-18.4°} \quad (mA).$$

When V_g is connected to the line, a voltage wave of an amplitude $\frac{Z_0}{Z_0+Z_g}V_g$ travels toward the load R_L, arriving with an amplitude $V_L^+ = \frac{Z_0 V_g}{Z_0 + Z_g} e^{-j\beta\ell}$, which causes a reflected wave with an amplitude $V_L^- = \Gamma_L V_L^+$. The reflected wave travels back toward the generator and is not reflected there because $Z_g = Z_0$, and $\Gamma_g = 0$.

$$\therefore\ V_L = \frac{Z_0 V_g}{Z_0 + Z_g} e^{-j\beta\ell}(1 + \Gamma_L) = \frac{1}{30} e^{-j\beta\ell} = 0.033\angle{-45°} \quad (V).$$

Similarly, $I_L = \frac{V_g}{Z_0 + Z_g} e^{-j\beta\ell}(1 - \Gamma_L) = \frac{4}{3} e^{-j\beta\ell} = 1.333\angle{-45°}$ (mA).

b) $S = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} = \frac{1 + \frac{1}{3}}{1 - \frac{1}{3}} = 2.$

c) $(P_{av})_L = \frac{1}{2}\mathcal{R}e(V_L I_L^*) = \frac{1}{2}\left(\frac{1}{30}\right)\left(\frac{4}{3}\times 10^{-3}\right) = 2.22\times 10^{-5}\,(W) = 0.022\,(mW)$

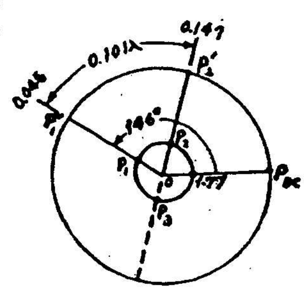
If $R_L = Z_0 = 50\,(\Omega)$, $\Gamma_L = 0$. $(P_{av})_L = \frac{V_g^2}{8Z_0} = 0.025\,(mW)$.
(matched condition)

P.8-20 $f = 2\times10^8$ (Hz), $\lambda = \frac{c}{f} = 1.5$ (m).

a) Open-circuited line, $\ell = 1$ (m), $\ell/\lambda = 0.667$.
Smith chart: Start from P_{oc} on the extreme right, rotate clockwise one complete revolution ($\Delta z' = \lambda/2$) and continue on for an additional 0.167λ to 0.417λ on the "wavelength toward generator" scale. Read $x = -j0.575$. $\longrightarrow$ $Z_i = 75\times(-j0.575) = -j43.1$ (Ω).
Draw a straight line from the $(0-j0.575)$ point through the center and intersect at $(0+j1.74)$ on the opposite side of the chart. $\longrightarrow$ $Y_i = \frac{1}{75}\times(j1.74) = j0.0232$ (S).

b) Short-circuited line, $\ell = 0.8$ (m), $\ell/\lambda = 0.533$.
Start from the extreme left point P_{sc}, rotate clockwise one complete revolution and continue on for an additional 0.033λ to read $x = j0.21 \longrightarrow Z_i = 75\times j0.21 = j15.8$ (Ω).
Draw a straight line from the $(0+j0.21)$ point through the center and intersect at $(0-j4.75)$ on the opposite side of the chart. $\longrightarrow$ $Y_i = \frac{1}{75}\times(-j4.75) = -j0.063$ (S).

P.8-21



$z_L = \frac{1}{50}(30+j10) = 0.6+j0.2$.

a) 1. Locate $z_L = 0.6+j0.2$ on Smith chart (Point P_1).
2. With center at O draw a $|\Gamma|$-circle through P_1, intersecting OP_{oc} at 1.77. $\longrightarrow$ $S = 1.77$.

b) $\Gamma = \frac{1.77-1}{1.77+1}e^{j146°} = 0.28e^{j146°}$.

c) 1. Draw line OP_1, intersecting the periphery at P_1'. Read 0.046 on "wavelengths toward generator" scale.
2. Move clockwise by 0.101λ to 0.147 (Point P_2').
3. Join O and P_2', intersecting the $|\Gamma|$-circle at P_2.
4. Read $z_i = 1+j0.59$ at P_2.

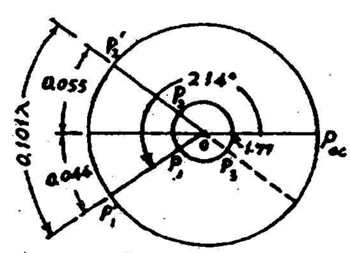
$$Z_i = 50z_i = 50+j29.5\ (\Omega).$$

d) Extend line $P_2'P_2O$ to P_3. Read $y_i = 0.75 - j0.43$.

$$Y_i = \frac{1}{50} y_i = 0.015 - j0.009 \text{ (S)}.$$

e) There is no voltage minimum on the line, but $V_L < V_i$.

P.8-22



$z_L = \frac{1}{50}(30 - j10) = 0.6 - j0.2$

a) Locate $z_L = 0.6 - j0.2$ on Smith chart (Point P_1). With center at O draw a $|\Gamma|$-circle through P_1, intersecting line OP_{sc} at 1.77. $\longrightarrow S = 1.77$.

b) $\Gamma = 0.28 e^{j214°}$

c) 1. Draw line OP_1, intersecting the periphery at P_1'. Read 0.454 on "wavelengths toward generator" scale.

2. Move clockwise by 0.101λ to 0.055 (Point P_2').

3. Join O and P_2', intersecting the $|\Gamma|$-circle at P_2.

4. Read $z_i = 0.61 + j0.23$ at P_2.

$$Z_i = 50 z_i = 30.5 + j11.5 \ (\Omega).$$

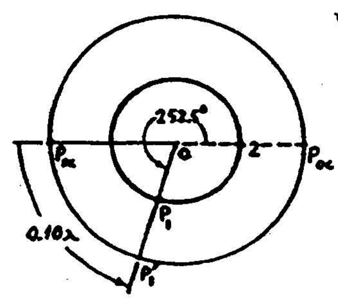
d) Extend line $P_2'P_2O$ to P_3. Read $y_i = 1.42 - j0.54$.

$$Y_i = \frac{1}{50} y_i = 0.0284 - j0.0108 \text{ (S)}.$$

e) There is a voltage minimum at $z_m' = 0.046\lambda$.

P.8-23 $\quad \lambda/2 = 25, \quad \lambda = 50$ (cm).

First voltage minimum occurs at $z_m' = \frac{5}{50} = 0.1\lambda$.



a) 1. Start from P_{sc} and rotate counterclockwise 0.10λ toward the load to P_1'.

2. Draw the $|\Gamma|$-circle, intersecting line OP_{oc} at 2 ($S = 2$).

3. Join OP_1', intersecting the $|\Gamma|$-circle at P_1.

4. Read $z_L = 0.675 - j0.475$.

$\longrightarrow Z_L = 50 z_L = 33.75 - j23.75$ (Ω).

b) $\Gamma = \frac{2-1}{2+1} e^{j\theta_r} = \frac{1}{3} e^{j252.5°}$.

c) If $Z_L = 0$, the first voltage minimum would be at $z'_m = \lambda/2 = 25$ (cm) from the short-circuit.

P. 8-24 $\quad f = 2\times10^8$ (Hz),

$\lambda = 1.5$ (m). $\longrightarrow \ell = \frac{\lambda}{4} = 0.375$(m).

Characteristic impedance of quarter-wire two-wire transmission line, $Z_0 = \sqrt{73\times300} = 148$ (Ω).

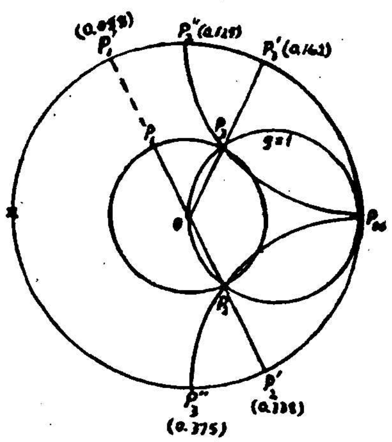
For a lossless air line, from Eqs. (8-23) and (8-24),

$$Z_0 = \sqrt{\frac{L}{C}} = \frac{1}{\pi}\sqrt{\frac{\mu_0}{\epsilon_0}}\cosh^{-1}\left(\frac{D}{2a}\right).$$

$$148 = 120\cosh^{-1}\left(\frac{D}{2a}\right).$$

Given $D = 2$ (cm) $\longrightarrow a$ (wire radius) $= 0.54$ (cm).

P. 8-25 $\quad z_L = (25+j25)/50 = 0.5+j0.5, \quad y_L = 1-j.$



a) See construction.

P_1: $z_L = 0.5 + j0.5$.

P_2: $y_L = 1 - j1 = y_{B1} \longrightarrow d_1 = 0$.

P_3: $y_{B2} = 1 + j1$.

$\longrightarrow d_2 = 0.162\lambda + (0.5 - 0.338)\lambda = 0.324\lambda$.

P_2'': $b_{B1} = j1 \longrightarrow \ell_1 = (0.25 + 0.125)\lambda = 0.375\lambda$.

P_3'': $b_{B2} = -j1 \longrightarrow \ell_2 = (0.375 - 0.25)\lambda = 0.125\lambda$.

P. 8-26 For $Z_0' = 75\,(\Omega) = 1.5\,Z_0$,
$Y_0' = \frac{1}{1.5} Y_0 = 0.667\,Y_0$. } Compared to Problem P. 8-25.

The required normalized stub admittances are

$$b_{B1}' = -b_{B2}' = \frac{j}{0.667} = j1.5.$$

The locations of points P_2'' and P_3'' are now different.

We have: $\ell_1' = 0.406\lambda$ and $\ell_2' = 0.094\lambda$.

There are no changes in the locations of the stubs:

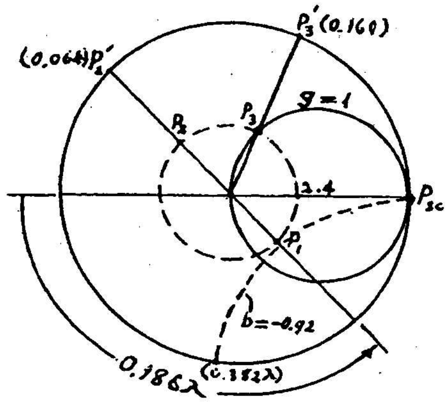
$$d_1' = d_1 = 0, \quad \text{and} \quad d_2' = d_2 = 0.324\lambda.$$

P. 8-27 Given: R_0 $75\,(\Omega)$, $S = 2.4$.

V_{min} at 0.335 (m) and 1.235 (m) from load.

$\longrightarrow \lambda = 2 \times (1.235 - 0.335) = 1.80\,(m)$.

First V_{min} at $\frac{0.335}{1.80} = 0.186\lambda$ from load.



Use a Smith chart.

1. Draw a centered circle (dashed) through $S=2.4$ point.
2. Locate point P_1 for z_L from V_{min} (point on extreme left) 0.186λ (clockwise) toward the load. $z_L = 1.39 - j0.98$.

a) $Z_L = 75\,z_L = 104.3 - j73.5\ (\Omega)$.

3. Locate the diametrically opposite point P_2 to find $y_L = 0.48 + j0.34$.

Read 0.064λ at point P_2'.

4. Use the Smith chart as an admittance chart and find the intersection of the $|\Gamma|$-circle with the $g=1$ circle at P_3: $y_B = 1 + j0.92$. Read 0.160λ at P_3'.

b) Location of stub $d = \underset{(\text{at } P_3')}{0.160\lambda} - \underset{(\text{at } P_2')}{0.064\lambda} = 0.096\lambda = 0.173\,(m)$.

Short-circuited stub length to give $b_B = -0.92$: $\ell = 0.382\lambda - 0.25\lambda$
$= 0.132\lambda = 0.238\,(m)$.

P.8-28 Use Smith chart as an impedance chart.——

Same construction as that in problem P.8-25, except point P_{sc} would be at the extreme left (marked by a ×) and the $g=1$ circle becomes a $r=1$ circle.

$$P_1: \quad z_L = (25+j25)/50 = 0.5+j0.5.$$

Two possible solutions:

At P_3: $z_{s3} = 1+j1$.

$$\longrightarrow d_3 = (0.162\lambda - 0.088\lambda) = 0.074\lambda.$$

At P_2: $z_{s3} = 1-j1$.

$$\longrightarrow d_2 = (0.338\lambda - 0.088\lambda) = 0.250\lambda.$$

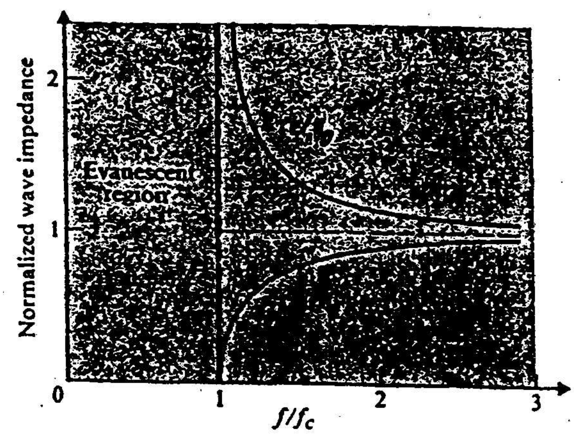
To achieve a match with a series stub having $R_0' = \frac{35}{50}R_0$, we need a normalized stub reactance $-j\frac{50}{35} = -j1.43$ for solution corresponding to P_3. From Smith chart we find the required stub length $\ell_3 = 0.347\lambda$.

Similarly for solution corresponding to P_2, a stub length with a normalized reactance $+j1.43$ is needed, which requires a stub length $\ell_2 = 0.153\lambda$.

Chapter 9

Waveguides and Resonators

P. 9-1 We use Eqs. (9-34) and (9-39) for Z_{TM} and Z_{TE} respectively. For air, $\eta = \eta_0 = 120\pi\,(\Omega) = 377\,(\Omega)$.



a) The normalized wave impedances are plotted as shown.

b) $Z_{TM} = \eta_0\sqrt{1-\left(\frac{f_c}{f}\right)^2}$,

$$Z_{TE} = \frac{\eta_0}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}}.$$

At $f = 1.1 f_c$, $\sqrt{1-\left(\frac{1}{1.1}\right)^2} = 0.417$.

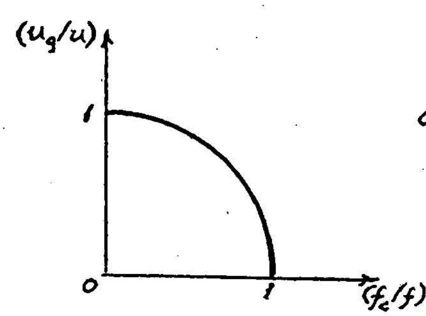
$Z_{TM} = 0.417\,\eta_0 = 157\,(\Omega)$.

$Z_{TE} = \frac{\eta_0}{0.417} = 904\,(\Omega)$.

At $f = 2.2 f_c$, $\sqrt{1-\left(\frac{f_c}{f}\right)^2} = \sqrt{1-\left(\frac{1}{2.2}\right)^2} = 0.891$.

$Z_{TM} = 0.891\,\eta_0 = 336\,(\Omega)$, $Z_{TE} = \frac{\eta_0}{0.891} = 423\,(\Omega)$.

P. 9-2 From Eq. (9-38), $\beta = k\sqrt{1-\left(\frac{f_c}{f}\right)^2} = \frac{\omega}{u}\sqrt{1-\left(\frac{f_c}{f}\right)^2}$.



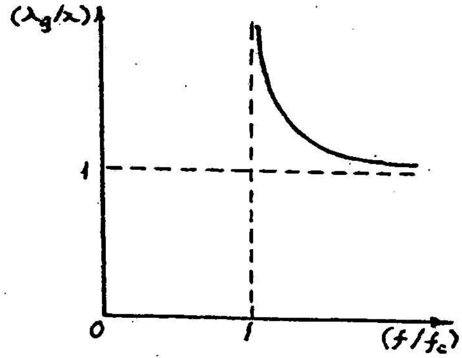
$$u_p = \frac{\omega}{\beta} = \frac{u}{\sqrt{1-(f_c/f)^2}}.$$

a) $u_g = \frac{1}{d\beta/d\omega} = u\sqrt{1-\left(\frac{f_c}{f}\right)^2}$.

$\longrightarrow \left(\frac{u_g}{u}\right)^2 + \left(\frac{f_c}{f}\right)^2 = 1$,

which indicates that the graph of (u_g/u) plotted versus (f_c/f) is a unit circle.

b) $\lambda_g = \dfrac{2\pi}{\beta} = \dfrac{2\pi u}{\omega}\dfrac{1}{\sqrt{1-(f_c/f)^2}} = \dfrac{\lambda}{\sqrt{1-(f_c/f)^2}}$.



$$\left(\frac{\lambda_g}{\lambda}\right)^2 = \frac{1}{1-(f_c/f)^2}.$$

$$\longrightarrow \left(\frac{\lambda_g}{\lambda}\right)^2 = \frac{(f/f_c)^2}{(f/f_c)^2-1}.$$

Graph shown on the left.

c) At $f/f_c = 1.25$,
$u_g/u = 0.60$, $\lambda_g/\lambda = 1.67$,
$u_p/u = 1.67$.

P.9-3 For TE waves between infinite parallel-plate waveguide in Fig. 9-3, we solve the following equation for $H_z^0(y)$: $\dfrac{d^2H_z^0(y)}{dy^2} + h^2H_z^0(y) = 0$,

with $H_z(y,z) = H_z^0(y)e^{-\gamma z}$. Boundary conditions to be satisfied at the conducting plates are:

$$\frac{dH_z^0(y)}{dy} = 0 \quad \text{at} \quad y=0 \text{ and } y=b.$$

a) Proper solution: $H_z^0(y) = B_n\cos\left(\dfrac{n\pi y}{b}\right)$; $\quad h = \dfrac{n\pi}{b}$, $n=1,2,3,\cdots$

b) From Eq. (9-26): $f_c = \dfrac{h}{2\pi\sqrt{\mu\epsilon}} = \dfrac{n}{2b\sqrt{\mu\epsilon}}$.

From TE_1 mode, $n=1$, $(f_c)_{TE_1} = \dfrac{u}{2b}$.

c) Instantaneous field expressions for TE_1 mode:

$$H_z(y,z;t) = B_1\cos\left(\frac{\pi y}{b}\right)\cos(\omega t-\beta_1 z), \quad \beta_1 = \omega\sqrt{\mu\epsilon}\sqrt{1-\left(\frac{f_c}{f}\right)^2}.$$

$$H_y(y,z;t) = -\frac{\beta_1 b}{\pi}B_1\sin\left(\frac{\pi y}{b}\right)\sin(\omega t-\beta_1 z).$$

$$E_x(y,z;t) = -\frac{\omega\mu b}{\pi}B_1\sin\left(\frac{\pi y}{b}\right)\sin(\omega t-\beta_1 z).$$

P.9-4 Parts (a) and (b) similar to Problem P. 9-3

a) Phase expressions for field components of TE modes:

$$H_z(y,z) = B_n \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta_n z},$$

$$\beta_n = \omega\sqrt{\mu\epsilon}\sqrt{1-\left(\frac{f_{cn}}{f}\right)^2}, \quad n = 1,2,3,\cdots$$

$$H_y(y,z) = \frac{j\beta_n b}{n\pi} B_n \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta_n z}.$$

$$E_x(y,z) = \frac{j\omega\mu b}{n\pi} B_n \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta_n z}.$$

b) From Eq. (9-26): $f_{cn} = \frac{n}{2b\sqrt{\mu\epsilon}}$.

c) Surface current densities: $\bar{J}_s = \bar{a}_n \times \bar{H}_t$.

On lower plate: $\bar{J}_{sl} = \bar{a}_y \times \bar{H}(0) = \bar{a}_x B_n e^{-j\beta_n z}$.

On upper plate: $\bar{J}_{su} = -\bar{a}_y \times \bar{H}(b) = \bar{a}_x (-1)^{n+1} B_n e^{-j\beta_n z}$

$$= \begin{cases} \bar{J}_{sl} \text{ for } n \text{ odd}, \\ -\bar{J}_{sl} \text{ for } n \text{ even}. \end{cases}$$

P.9-5 a) $\lambda_g = 2 \times 2.65 = 5.30$ (cm).

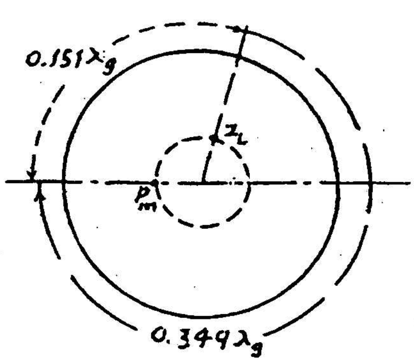
For TE_{10} mode: $f_c = \frac{c}{2a} = \frac{3\times 10^8}{2\times 0.025} = 6\times 10^9$ (Hz).

$\lambda_c = 2a = 2\times 0.025 = 0.05$ (m).

From Eqs. (9-30) and (9-31): $\left(\frac{c}{\lambda_g}\right)^2 = f^2 - f_c^2$

$$\longrightarrow f = \sqrt{f_c^2 + \left(\frac{c}{\lambda_g}\right)^2} = \sqrt{6^2 + \left(\frac{0.3}{0.053}\right)^2} \times 10^9 = 8.25\times 10^9 \text{(Hz)} = 8.25 \text{ (GHz)}.$$

b) Use Smith chart. Draw $|\Gamma| = \frac{2-1}{2+1} = \frac{1}{3}$ circle through $S=2$ point.



Shifting V_m toward load by $\frac{0.80}{3.30} = 0.151\lambda_g$ places point P_m from the load $(0.5-0.151)\lambda_g = 0.349\lambda_g$ toward the generator.

Read $z_L = 0.99 + j0.71$.

$$Z_{TE_{10}} = \frac{\eta_0}{\sqrt{1-(f_c/f)^2}} = \frac{377}{\sqrt{1-(6/8.25)^2}} = 549 (\Omega)$$

$$\longrightarrow Z_L = (0.99 + j0.71)\times 549 = 544 + j390 \text{ } (\Omega).$$

c) $P_{load} = 10\left(1 - \frac{1}{3^2}\right) = 8.89$ (W).

P.9-6 TM_{11} mode in air-filled rectangular waveguide operating at angular frequency $\omega = 2\pi f$ (see Eq. 9-65):

a) $$E_z^0(x,y) = E_0 \sin\left(\frac{\pi x}{a}\right)\sin\left(\frac{\pi y}{b}\right).$$

Setting $H_z^0 = 0$ in Eqs. (9-11) through (9-14):

$$H_x^0(x,y) = \frac{j\omega\epsilon}{h_{11}^2}\frac{\partial E_z^0}{\partial y} = \frac{j\omega\epsilon}{h_{11}^2}\left(\frac{\pi}{b}\right)E_0 \sin\left(\frac{\pi x}{a}\right)\cos\left(\frac{\pi y}{b}\right),$$

$$H_y^0(x,y) = -\frac{j\omega\epsilon}{h_{11}^2}\frac{\partial E_z^0}{\partial x} = -\frac{j\omega\epsilon}{h_{11}^2}\left(\frac{\pi}{a}\right)E_0 \cos\left(\frac{\pi x}{a}\right)\sin\left(\frac{\pi y}{b}\right),$$

$$E_x^0(x,y) = -\frac{j\beta_{11}}{h_{11}^2}\frac{\partial E_z^0}{\partial x} = -\frac{j\beta_{11}}{h^2}\left(\frac{\pi}{a}\right)E_0 \cos\left(\frac{\pi x}{a}\right)\sin\left(\frac{\pi y}{b}\right),$$

$$E_y^0(x,y) = -\frac{j\beta_{11}}{h_{11}^2}\frac{\partial E_z^0}{\partial y} = -\frac{j\beta_{11}}{h_{11}^2}\left(\frac{\pi}{b}\right)E_0 \sin\left(\frac{\pi x}{a}\right)\cos\left(\frac{\pi y}{b}\right),$$

where $h_{11}^2 = \left(\frac{\pi}{a}\right)^2 + \left(\frac{\pi}{b}\right)^2$, $\beta_{11} = \sqrt{\omega^2\mu\epsilon - h_{11}^2}$. Variations in z-direction are described by the factor $e^{-j\beta_{11}z}$.

b) From Eq. (9-26), $(f_c)_{TM_{11}} = \frac{h_{11}}{2\pi}c = \frac{c}{2}\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$.

$$(\lambda_c)_{TM_{11}} = \frac{c}{(f_c)_{TM_{11}}} = \frac{2}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}} = \frac{2ab}{\sqrt{a^2+b^2}}.$$

$$\lambda_g = \frac{2\pi}{\beta_{11}} = \frac{\lambda}{\sqrt{1-(f_c/f)^2}} = \frac{c}{\sqrt{f^2 - f_c^2}}.$$

P.9-7 Rectangular waveguide: $a = 7.21$ (cm), $b = 3.40$ (cm).

Eq. (9-69): $$(\lambda_c)_{mn} = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}.$$

Modes with the shortest $\lambda_c < 5$ (cm) are:

Mode	TE_{10}	TE_{20}	TE_{01}	TE_{11}/TM_{11}
λ (cm)	14.4	7.20	6.80	6.15

a) For $\lambda = 10$ (cm), the only propagating mode is TE_{10}.

b) For $\lambda = 5$ (cm), the propagating modes are: TE_{10}, TE_{20}, TE_{01}, TE_{11}, and TM_{11}.

P. 9-8 $\quad (f_c)_{mn} = \frac{1}{2\sqrt{\mu\epsilon}}\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2} = \frac{1}{2a\sqrt{\mu\epsilon}}F(m,n).$

a) $a=2b$, $F(m,n)=\sqrt{m^2+4n^2}$

Modes	F(m,n)
TE_{10}	1
TE_{01}, TE_{20}	2
TE_{11}, TM_{11}	$\sqrt{5}$
TE_{02}	4
TM_{12}	$\sqrt{17}$
TM_{22}	$\sqrt{20}$

b) $a=b$, $F(m,n)=\sqrt{m^2+n^2}$.

Modes	F(m,n)
TE_{10}, TE_{01}	1
TE_{11}, TM_{11}	$\sqrt{2}$
TE_{02}, TE_{20}	2
TM_{12}	$\sqrt{5}$
TM_{22}	$2\sqrt{2}$

P. 9-9 $\quad f = 3\times 10^9$ (Hz), $\lambda = c/f = 0.1$ (m).

Let $a=kb$, $1<k<2$. $(f_c)_{mn} = \frac{3\times 10^8}{2a}\sqrt{m^2+k^2n^2}$.

a) $(f_c)_{10} = \frac{1.5\times 10^8}{a}$ for the dominant TE_{10} mode.

For $f > 1.2\,(f_c)_{10}$: $\quad a > 0.06$ (m).

The next higher-order mode is TE_{01} with $(f_c)_{01} = \frac{1.5\times 10^8}{b}$.

For $f < 0.8\,(f_c)_{01}$: $\quad b < 0.04$ (m).

We choose $a = 6.5$ (cm) and $b = 3.5$ (cm).

b) $u_p = \frac{c}{\sqrt{1-(\lambda/2a)^2}} = 4.70\times 10^8$ (m/s),

$\lambda_g = \frac{\lambda}{\sqrt{1-(\lambda/2a)^2}} = 0.157$ (m) $= 15.7$ (cm),

$\beta = \frac{2\pi}{\lambda_g} = 40.1$ (rad/m),

$(Z_{TE})_{10} = \frac{\eta_0}{\sqrt{1-(\lambda/2a)^2}} = 590$ (Ω).

P. 9-10 Given: $a=2.5\times10^{-2}$ (m), $b=1.5\times10^{-2}$(m), $f=7.5\times10^{9}$(Hz).

a) $\lambda=\frac{c}{f}=\frac{3\times10^{8}}{7.5\times10^{9}}=0.04$ (m),

$F_1=\sqrt{1-(\lambda/2a)^2}=0.60$,

$\lambda_g=\lambda/F_1=0.0667$ (m) $=6.67$ (cm),

$\beta=2\pi/\lambda_g=94.2$ (rad/m),

$u_p=c/F_1=5\times10^{8}$ (m/s),

$u_g=c\cdot F_1=1.8\times10^{8}$ (m/s),

$(Z_{TE})_{10}=\eta_0/F_1=200\pi=628$ (Ω).

b) $\lambda'=\frac{u}{f}=\frac{\lambda}{\sqrt{2}}=0.0283$ (m),

$F_2=\sqrt{1-(\lambda'/2a)^2}=0.825$,

$\lambda_g'=\lambda'/F_2=0.0343$ (m) $=3.43$ (cm),

$\beta'=2\pi/\lambda_g'=183.2$ (rad/m),

$u_p'=u/F_2=2.57\times10^{8}$ (m/s),

$u_g'=u\cdot F_2=1.75\times10^{8}$ (m/s),

$(Z_{TE})_{10}=\frac{\eta_0}{\sqrt{2}\,F_2}=323$ (Ω).

P. 9-11 Part (a) has been done in problem P. 9-6, part (a).

b) Use Eq. (7-79) to find the average power transmitted along the waveguide.

$$P_{av}=\frac{1}{2}\int_0^b\int_0^a\left[E_x^0H_y^0-E_y^0H_x^0\right]dx\,dy$$

$$=\frac{\omega\epsilon\beta_{11}E_0^2ab}{8\left[\left(\frac{\pi}{a}\right)^2-\left(\frac{\pi}{b}\right)^2\right]}.$$

P. 9-12 a) $E_z(x,y,z;t)=E_0\sin(100\pi x)\sin(100\pi y)\cos(2\pi10^{10}t-\beta z)$

$$=E_0\sin\left(\frac{2\pi}{a}x\right)\sin\left(\frac{\pi}{b}y\right)\cos\left(2\pi10^{10}t-\beta z\right).$$

Mode of operation: TM_{21}. $\omega=2\pi f=2\pi10^{10}$(rad/s).

b) $(f_c)_{21} = \frac{c}{2}\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2} = \frac{3\times10^8}{2}\sqrt{\left(\frac{2}{0.05}\right)^2+\left(\frac{1}{0.025}\right)^2}$

$= 8.48\times10^9$ (Hz).

$\beta = \frac{\omega}{c}\sqrt{1-\left(\frac{f_c}{f}\right)^2} = \frac{2\pi 10^{10}}{3\times10^8}\sqrt{1-\left(\frac{8.48}{10}\right)^2} = 111$ (rad/m).

Eq. (9-34): $(Z_{TM})_{21} = \eta_0\sqrt{1-\left(\frac{f_c}{f}\right)^2} = 377\sqrt{1-\left(\frac{8.48}{10}\right)^2}$

$= 377\times0.53 = 200\ (\Omega)$.

$\lambda_g = \frac{\lambda}{\sqrt{1-(f_c/f)^2}} = \frac{3\times10^8}{10^{10}\times0.53} = 0.0566\ (m) = 5.66\ (cm)$.

P. 9-13 TE mode in 0.025 (m) × 0.025 (m) air-filled square waveguide:

$$H_z(x,y,z;t) = H_0\cos\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)\cos(\omega t-\beta z)$$
$$= 0.3\cos(80\pi y)\cos(\omega t - 280z).$$

a) $\frac{n\pi}{b} = 80\pi = \frac{2\pi}{0.025} \longrightarrow n=2;\ m=0.$

$\longrightarrow TE_{02}$ mode.

b) From Eq. (9-68):

$(f_c)_{02} = \frac{c}{2}\,\frac{2}{b} = \frac{c}{b} = \frac{3\times10^8}{0.025} = 1.2\times10^{10}\ (Hz) = 12\ (GHz)$.

From Eq. (9-38): $\beta = \frac{\omega}{c}\sqrt{1-\left(\frac{f_c}{f}\right)^2} = \frac{2\pi}{c}\sqrt{f^2-f_c^2}$.

$f = \sqrt{\left(\frac{\beta c}{2\pi}\right)^2+f_c^2} = \sqrt{\left(\frac{280\times3\times10^8}{2\pi}\right)^2+(1.2\times10^{10})^2} = 1.8\times10^{10}\ (Hz)$
$= 18\ (GHz)$.

$Z_{TE} = \frac{\eta_0}{\sqrt{1-(f_c/f)^2}} = \frac{377}{\sqrt{1-(1.2/1.8)^2}} = 506\ (\Omega)$,

$\lambda_g = \lambda/\sqrt{1-(f_c/f)^2} = c/f\sqrt{1-(f_c/f)^2} = 2.24\times10^{-2}\ (m) = 2.24\ (cm)$.

c) $P_{av} = \frac{1}{2}\int_0^b\int_0^b \frac{|E_x|^2}{2Z_{TE}}\,dx\,dy = \frac{\omega^2\mu_0^2H_0^2}{4Z_{TE}}\int_0^b \sin^2\left(\frac{2\pi}{b}x\right)dx$

$= \frac{(2\pi f)^2\mu_0^2H_0^2}{4Z_{TE}}\left(\frac{b^2}{2}\right) = 280\ (W)$.

P. 9-14 Substituting Eq. (9-97) in Eq. (9-24):

a) $$\gamma = j\left[\omega^2\mu\epsilon\left(1-j\frac{\sigma_d}{\omega\epsilon}\right)-h^2\right]^{1/2}$$
$$= j\sqrt{\omega^2\mu\epsilon-h^2}\left\{1-j\omega\mu\sigma_d(\omega^2\mu\epsilon-h^2)^{-1}\right\}^{1/2}$$
$$\cong j\sqrt{\omega^2\mu\epsilon-h^2}\left\{1-\frac{j\omega\mu\sigma_d}{2}(\omega^2\mu\epsilon-h^2)^{-1}\right\}.$$

From Eq. (9-28), $\sqrt{\omega^2\mu\epsilon-h^2} = \omega\sqrt{\mu\epsilon}\sqrt{1-(f_c/f)^2}$.

Hence, $$\gamma = \alpha_d + j\beta,$$

with $$\alpha_d = \frac{\sigma_d}{2}\sqrt{\frac{\mu}{\epsilon}}\frac{1}{\sqrt{1-(f_c/f)^2}} = \frac{\sigma_d\eta}{2\sqrt{1-(f_c/f)^2}}.$$

b) At $f = 4\times10^9$ (Hz), TE_{10} is the only propagating mode which has a cutoff frequency of

$$(f_c)_{TE_{10}} = \frac{u}{2a} = \frac{c/\sqrt{\epsilon_r}}{2a} = \frac{3\times10^8/\sqrt{4}}{2\times0.025} = 3\times10^9\ (\text{Hz}).$$

Thus, $$\alpha_d = \frac{3\times10^{-5}\times377}{2\sqrt{1-(3/4)^2}} = 0.0085\ (\text{Np/m}) = 0.074\ (\text{dB/m}).$$

P. 9-15 $(f_c)_{mn} = \frac{c}{2}\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2}$.

a) $(f_c)_{10} = \frac{c}{2a} = \frac{3\times10^8}{2\times0.025} = 6\times10^9\ (\text{Hz}) = 6\ (\text{GHz})$.

Next higher mode:

	$b=0.25a$	$b=0.50a$	$b=0.75a$	
$(f_c)_{01} = \frac{c}{2b}$	24	12	8	(GHz)
$(f_c)_{11} = \frac{c}{2a}\sqrt{1+\left(\frac{a}{b}\right)^2}$	24.7	13.4	10	(GHz)
$(f_c)_{20} = \frac{c}{a}$	12	12	12	(GHz)
	0.85×12	0.85×12	0.85×8	
Usable bandwidth is from $1.15\times6 = 6.9$ (GHz) to:	10.2	10.2	6.8	(GHz)
Permissible bandwidth:	3.3 (GHz)	3.3 (GHz)	None <6.9	

b) From Eq. (9-101): $P_{av} = \frac{E_0^2 ab}{4\eta_0}\sqrt{1-\left(\frac{f_c}{f}\right)^2} = 21.34\left(\frac{b}{a}\right)$.

$\longrightarrow$ $P_{av} = 5.3$ (W) for $b=0.25a$, and 10.7 (W) for $b=0.50a$.